



**COMSATS University Islamabad,
Sahiwal Campus**

Flex Work

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Bachelor of Science in Computer Science (2022-2026)

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DECLARATION

We hereby declare that this App (Flex-Work), neither whole nor as a part has been copied out from any source. It is further declared that we have developed this App and accompanied report entirely based on our personal efforts. If any part of this project is proved to be copied out from any source or found to be reproduction of some other. We will stand by the consequences. No Portion of the work presented has been submitted of any application for any other degree or qualification of this or any other university or institute of learning.

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CERTIFICATE OF APPROVAL

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Executive Summary

Flex Work is an Android-based mobile application designed to bridge the growing gap between non-professional job seekers and individuals or businesses seeking flexible, short-term, and daily-wage workers. Built using Kotlin and powered by Firebase's real-time backend, the platform provides a seamless, efficient, and secure environment where job seekers can discover opportunities based on their availability and skill level, while employers can post, manage, and track job listings with ease. The core purpose of Flex Work is to empower unskilled and semi-skilled workers such as helpers, loaders, cleaners, delivery assistants, and daily laborers by giving them access to real employment opportunities within their locality through a simple and accessible digital platform. It addresses the critical barriers that currently exist in Pakistan's informal labor economy, where transparency, reliability, and structured job-matching systems are largely absent.

The application offers a multi-role ecosystem consisting of Job Seekers, Employers, and Admin. Job Seekers can register, browse job listings, apply for work, view application status, receive payments, and maintain a profile with ratings and reviews that help build credibility over time. Employers can post job openings, manage applicants, communicate through in-app messaging, process secure payments, and review workers after job completion. The admin oversees the entire ecosystem by moderating user activity, managing job posts, resolving issues, and ensuring the platform operates smoothly and is free from misuse. Using Firebase Authentication, Firebase Storage, and Cloud Messaging, the system ensures real-time synchronization, high availability, and secure data handling for all users.

Flex Work was developed using an Agile and Iterative methodology, allowing the development team to continuously refine features through feedback, incremental builds, and rapid testing cycles. The project encompasses requirements engineering via detailed SRS documentation, architectural planning through system and process flow diagrams, and thorough design modeling using UML diagrams such as sequence, use case, and class diagrams. The system architecture follows a client-server model, where the Android application serves as the client, while Firebase handles backend services including authentication, real-time database management, storage, messaging, and analytics. This architecture ensures high scalability, enabling the platform to support an increasing user base and additional features without compromising performance.

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Abbreviations

FCM	Firestore Cloud Messaging
SDLC	Software Development Life Cycle
UC	User Case
UI	User Interface
FT	Functional Testing
UML	Unified Modeling Language
FAQ	Frequently Asked Questions

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Chapter 1

Introduction

1 Introduction

The rapid rise in digital employment platforms has transformed how individuals search for work, yet a significant segment of the workforce non-professional and unskilled laborers continues to face barriers in accessing flexible and short-term job opportunities. Many existing job portals focus primarily on skilled professions, corporate hiring, or full-time employment, leaving daily-wage workers, helpers, loaders, cleaners, and similar individuals without a structured digital platform tailored to their needs. Flex Work addresses this gap by providing an accessible, intuitive, and real-time job-matching application designed specifically for unskilled and semi-skilled job seekers and employers seeking immediate or short-term assistance. Built for Android devices and leveraging Firebase's cloud infrastructure, the system combines usability, transparency, and efficiency to streamline the hiring process in a way that aligns with the realities of Pakistan's informal labor market.

Flex Work enables two primary user roles Job Seekers and Employers to interact in a secure and well-structured environment. Job Seekers can register, search for work based on location and category, view job details, apply with a single tap, receive updates instantly, communicate directly with employers, and obtain secure in-app payments. Employers can post job listings, specify requirements, review applicants, process payments, and evaluate workers through ratings and reviews. The system further incorporates an Admin role responsible for monitoring job postings, validating user activity, resolving conflicts, and ensuring the platform remains fair, transparent, and aligned with community guidelines. Through this multi-role structure, Flex Work establishes a balanced and trust-driven ecosystem that supports both ends of the employment spectrum.

Technologically, Flex Work is developed using Kotlin a modern language designed for safety, performance, and seamless Android integration. Firebase serves as the backend, offering real-time database management, authentication, cloud messaging, storage, analytics, and a secure foundation for data operations.

1.1 Brief Overview

Flex Work is an Android-based mobile application designed to provide a seamless digital platform for connecting non-professional job seekers with short-term, temporary, and daily-wage employment opportunities. The system targets individuals who may not possess specialized skills or formal qualifications but are actively looking for flexible work such as

loading and unloading tasks, cleaning assistance, house shifting support, delivery services, and general labor. The app focuses on simplicity, accessibility, and real-time interaction, enabling job seekers and employers to engage through an easy-to-use interface optimized for Android devices. By integrating features such as job posting, job search, in-app communication, application tracking, and secure payment handling, Flex Work aims to streamline the entire hiring cycle within a single unified platform.

Built using Kotlin and supported by Firebase's real-time backend services, Flex Work ensures reliable performance, swift updates, and low latency. The use of Firebase Authentication, Firebase Storage, and Cloud Messaging allows the app to manage user accounts securely, sync data instantaneously, and deliver real-time notifications for job alerts, application updates, and payment statuses. Employers can create job listings with details such as location, description, duration, and salary, while job seekers can browse these listings using filters like job category, distance, and pay range.

The platform operates with a three-role ecosystem: Job Seeker, Employer, and Admin. Job Seekers can register, maintain profiles, apply for jobs, receive notifications, and manage payments securely. Employers have access to job posting tools, applicant review features, chat messaging, and job assignment functions. The admin role is responsible for moderating content, handling user issues, maintaining platform integrity, and ensuring compliance with system rules and community guidelines. This role-based structure ensures that Flex Work remains organized, reliable, and secure for all users.

As a whole, Flex Work serves as a scalable and user-centric solution aimed at modernizing the informal job market. By offering a digital bridge between job seekers and employers, the app enhances accessibility, reduces unemployment barriers, and promotes trust through ratings, reviews, and secure communication channels. Its lightweight design, real-time responsiveness, and community-driven features align Flex Work with the practical needs of short-term labor markets, making it a promising platform for continuous growth and future enhancements.

1.2 Relevance to Course Modules

The Flex Work application directly reflects the practical implementation of several course modules studied throughout the Computer Science program. Concepts from Software Engineering, Mobile Application Development, Human-Computer Interaction, Database Systems, and Cloud Computing were applied during the development of this project. The system integrates theoretical knowledge with hands-on development practices, demonstrating

how academic learning can be utilized to build a real-world Android application that solves a meaningful societal problem.

- **Reliable and Scalable Software Applications**

Flex Work demonstrates the principles of reliability and scalability taught in software engineering modules. Reliability is achieved through Firebase Authentication, real-time data synchronization, and secure backend operations that allow job posts, applications, and messages to be updated instantly. Scalability is supported by Firebase's cloud infrastructure, which automatically manages increased user traffic, expanding job listings, and growing data storage demands. Modular coding practices in Kotlin make it easier to integrate future functionalities enhanced filtering, or employer verification without disrupting existing systems.

- **Training and Executing Flex Work**

Building Flex Work required continuous learning, experimentation, and iterative development based on Agile methodology. The development team practiced essential skills such as Android UI design, Kotlin programming, Firebase integration, and cloud messaging implementation. Early prototypes focused on user registration, job posting workflows, and application submission. These prototypes were refined through repeated testing, debugging, and interface improvements. Tools such as Android Studio, Firebase Console, GitHub, and diagrams.net were used to train the developers in industry-standard practices. Through regular sprint cycles, the team built key modules authentication, job management, payment processing, messaging, and admin moderation ensuring smooth integration, optimized performance, and maintainable code structure.

- **Real-Time Notifications**

Real-time notifications are an essential component of the Flex Work application, enhancing user engagement and ensuring timely communication. Powered by Firebase Cloud Messaging (FCM), these notifications alert users about new job postings, application approvals or rejections, payment confirmations, and chat messages. Notifications appear instantly within the Android interface without interrupting the user experience. In future updates, the notification system will also support employer reminders, job expiry alerts, and personalized recommendations. This module aligns with networking and mobile computing course concepts involving asynchronous communication, cloud services, and event-driven systems.

1.3 Project Background

Flex Work is an Android-based job-matching platform developed to address the growing demand for flexible, short-term, and daily-wage employment opportunities among non-professional workers in Pakistan. While digital job portals exist, most of them primarily focus on skilled professionals and corporate-level careers, leaving a considerable gap for individuals who rely on manual labor, part-time tasks, or day-to-day work for survival. Many workers in this segment struggle to find trustworthy employers, secure payments, or access transparent information regarding job availability. Similarly, employers often face difficulties when searching for honest, reliable, and readily available helpers for tasks such as house shifting, cleaning, loading, delivery support, and general labor. These challenges highlighted the need for a dedicated, accessible, and technology-driven platform designed specifically for this workforce.

The concept of Flex Work originated from the observation that the informal labor market lacks digital structure. Traditional hiring methods word-of-mouth referrals, local contacts, or in-person searches are often unreliable, time-consuming, and unsafe for both parties. By creating a mobile application that directly connects job seekers and employers, the development team aimed to provide a modernized, real-time solution that simplifies the entire hiring lifecycle. Flex Work ensures that job seekers can view available jobs instantly, apply with a single tap, communicate with employers, and receive timely payments upon task completion. Employers, on the other hand, gain access to a pool of verified workers, allowing them to post jobs, track applications, and manage payments through a secure, streamlined interface.

Technically, the project leverages Kotlin for Android development and Firebase for backend services, chosen for their performance, reliability, and real-time capabilities. Firebase Authentication manages user verification, Firebase Storage stores job-related data, and Firebase Cloud Messaging delivers instant notifications about job alerts, application updates, and system-level changes. These technologies form the foundation of a system that responds quickly to user actions and supports dynamic interactions between job seekers and employers. The app's architecture supports scalability, ensuring performance remains stable even as the user base and job volume increase.

1.4 Methodology and Software Development Lifecycle (SDLC)

The development of Flex Work followed a structured Software Development Lifecycle (SDLC) to ensure that the system was built in a systematic, scalable, and user-focused manner. Since

the project involved evolving requirements and continuous refinement based on testing, the Agile Methodology combined with an Iterative development model was selected. This allowed the development team to gradually build features, incorporate feedback, and improve the system step by step. Each phase from requirements gathering to implementation and testing was documented clearly, ensuring that the final product aligned closely with the needs of job seekers, employers, and administrators. [1]

1.4.1 Requirement Analysis

The first stage of development involved extensive requirement analysis to understand the needs of both stakeholders and end-users. Discussions, real-world observations, and initial research formed the foundation for the system's core functionalities. The team analyzed the challenges faced by non-professional job seekers and employers in the informal labor market and translated these findings into structured functional and non-functional requirements.

These requirements including registration, job posting, job searching, real-time notifications, secure payments, and admin moderation were formally documented in the Software Requirements Specification (SRS). The SRS served as the primary reference for all subsequent phases of the SDLC.

1.4.2 System and Design Modeling

After defining the requirements, the team moved into the design modeling phase. Several UML diagrams were constructed to visualize system behavior, data flow, and component interactions.

Key design artifacts included:

- Use Case Diagrams to show interactions between Job Seeker, Employer, and Admin.
- Sequence Diagrams to illustrate workflows such as login, job posting, applying for jobs, and payment processing.
- Class Diagrams to define the structure of objects such as Users, Jobs, Applications, Payments, and Messages.
- Architectural Diagrams to represent the client-server model with Firebase backend.
- These models ensured that the system's structure remained consistent, maintainable, and aligned with backend requirements.

1.4.3 Design and Architecture

Flex Work was designed using modular architecture, allowing each core component authentication, job management, messaging, notifications, payments, and admin controls to operate independently yet cohesively.

The backend architecture relies on Firebase services including Firestore, Firebase Authentication, Cloud Storage, and Cloud Messaging. This serverless architecture supports real-time updates, scalability, and fast data synchronization.

The frontend design followed Android UI/UX principles to ensure usability for individuals with minimal technical literacy, including clean layouts, readable fonts, intuitive navigation paths, and task-oriented screens.

1.4.4 Implementation

The implementation phase involved the actual coding and integration of modules using Agile sprints.

Key technologies included:

- Kotlin for developing Android screens, logic, and system interactions.
- Firebase Authentication for secure login and registration.
- Firestore for managing jobs, applications, users, and payments.
- Firebase Cloud Messaging (FCM) for real-time notifications.
- Stripe/PayPal SDKs for processing job payments securely.

Each sprint focused on implementing one major feature starting with login and registration, followed by job posting, job searching, application tracking, real-time messaging, payment handling, and admin tools.

1.4.5 Testing

Testing was conducted throughout the development process to ensure accuracy, stability, and performance.

The team conducted:

- Functional Testing (login, registration, job posting, job search, payments)
- UI/UX Testing to verify ease of navigation and readability

- Integration Testing to ensure modules worked together without conflicts
- Performance Testing for response time, database queries, and loading speed
- Security Testing for authentication, data access, and transaction safety
- Firebase Crashlytics assisted in monitoring crashes, errors, and device-specific issues.

1.4.6 Deployment

Following testing and refinement, the application was packaged into a release-ready APK and deployed for internal evaluation and academic submission. Firebase hosting ensured backend availability, while Android Studio was used for generating final builds. Deployment included version control, build configuration, and environment setup for stable performance.

1.4.7 Maintenance and Future Enhancements

Maintenance is an ongoing process intended to improve system performance, fix bugs, and integrate new features.

Planned enhancements include:

- Advanced employer analytics
- Wallet system for storing earnings
- Improved dispute resolution mechanisms

Chapter 2

Problem Definition

2 Problem Definition

In today's rapidly evolving digital landscape, numerous platforms exist for professional employment, skilled labor recruitment, and corporate hiring. However, there remains a significant gap in systems designed specifically for non-professional, unskilled, and daily-wage workers. These individuals often rely on informal and unreliable methods such as word-of-mouth referrals, neighborhood contacts, or manual searching to secure temporary work. As a result, they face challenges such as inconsistent income, lack of access to genuine job opportunities, exploitation, unsafe work conditions, and delayed payments. At the same time, employers seeking helpers for short tasks such as house shifting, cleaning, delivery assistance, or manual labor find it difficult to quickly find trustworthy workers when needed.

Additionally, the informal job marketplace lacks transparency, verification mechanisms, and structured communication channels, which makes it difficult for both employers and workers to establish trust. Many job seekers do not have the technological literacy to navigate complex job portals, and therefore require a system with a simple, accessible, and intuitive interface. Similarly, employers require a reliable platform where they can post job requirements, evaluate applicants, and process payments securely.

Another problem is the absence of a centralized system that supports real-time job posting, application tracking, and secure payments. Existing labor solutions fail to address the needs of the unskilled workforce, leaving a large portion of the population disconnected from digital employment opportunities. There is also a lack of admin-level oversight to monitor platform activity, moderate user behavior, and ensure fairness within the ecosystem.

Flex Work aims to solve these challenges by providing a dedicated Android-based platform that connects job seekers and employers in real time. The system offers features such as job posting, job searching, application submission, secure digital payments, in-app communication, and user ratings. By digitizing the informal labor market, Flex Work reduces barriers to employment, enhances transparency, and supports workers who rely on task-based income. The project focuses on building a safe, reliable, and accessible ecosystem that empowers unskilled laborers while simplifying the hiring process for employers. [2]

2.1 Problem Statement

The current job marketplace lacks a dedicated digital platform that effectively connects non-professional, unskilled, and daily-wage workers with employers seeking immediate or short-

term labor. Existing job portals are primarily designed for skilled professionals and formal employment, leaving a significant population without accessible or reliable means to find work. As a result, job seekers often depend on informal networks, physical searching, or unreliable intermediaries, which leads to inconsistent job availability, lack of transparency, limited trust, and frequent exploitation.

Employers face similar challenges when attempting to hire workers for temporary tasks such as household shifting, cleaning, loading or unloading, small repairs, or general labor assistance. Without a structured system, employers struggle to identify available workers, verify their reliability, communicate efficiently, or ensure secure payment after job completion.

The absence of a centralized, real-time, and user-friendly job-matching system creates multiple **issues:**

- No reliable medium for unskilled workers to find flexible jobs
- Limited visibility of available workers for employers
- Lack of structured communication, leading to misunderstandings and delays
- No secure digital payment mechanism, resulting in disputes or delayed compensation
- No formal user verification, leading to safety and trust concerns
- No admin oversight, making platforms vulnerable to misuse, fraud, and misinformation

These challenges necessitate a dedicated mobile application that provides immediate access to job opportunities, simplifies employer worker interactions, ensures secure transactions, and maintains platform integrity through transparent processes and administrative monitoring.

2.2 Deliverables and Development Requirements

The development of Flex Work involves a set of essential deliverables that ensure the system is properly designed, implemented, and evaluated. These deliverables include the complete Android application, the Software Requirements Specification (SRS), the Software Design Description (SDD), UML diagrams, testing documentation, and the final project report and presentation. The project also requires a functional user interface, a Firebase-based backend, payment module integration, and real-time notification services. Development requirements include Android Studio, Kotlin, Firebase services, third-party SDKs, and proper testing tools to validate system functionality, performance, and security. These deliverables and requirements collectively ensure that Flex Work is scalable, reliable, and ready for deployment and academic evaluation.

2.2.1 User Interface (UI) Design

The User Interface for Flex Work is designed to be simple, intuitive, and accessible for non-technical users such as daily-wage workers and small employers. The layout focuses on clarity, minimal navigation steps, and easy interaction with essential features like job search, job posting, and communication. The UI uses clean visual elements, readable typography, and clearly labeled buttons to ensure users can quickly understand how to perform tasks. Key screens such as login, registration, job listings, job details, applications, employer dashboard, and payment confirmation are organized to reduce confusion and guide the user smoothly through each process. Overall, the UI design ensures a user-friendly experience that supports the application's goal of providing quick and efficient access to flexible job opportunities.

2.2.2 Backend Development

The backend of Flex Work is built using Firebase to support real-time data management, secure authentication, and scalable cloud storage. Firebase Authentication handles user login and verification, while Firestore stores all system data, including job posts, applications, user profiles, and notifications. Real-time synchronization ensures that job listings, messages, and application updates are instantly reflected on all devices. The backend also integrates local mobile payment solutions such as Jazz Cash and EasyPaisa, enabling secure and convenient transactions for users within Pakistan. This cloud-driven architecture minimizes server overhead, ensures high availability, and provides a stable foundation for the application's core features.

2.2.3 Third-Party Service Integration

Flex Work relies on several third-party services to improve system reliability, enhance user experience, and streamline core operations. Firebase Cloud Messaging enables real-time notifications, ensuring that job seekers and employers instantly receive alerts for new job postings, application updates, messages, and important system events. Firebase Authentication is used to securely verify user identities through email or phone numbers, reducing the risk of unauthorized access. The platform also integrates Jazz Cash and EasyPaisa, two of Pakistan's most widely used mobile payment services, allowing users to perform secure and convenient transactions directly through the app. These integrations eliminate the need for complex backend infrastructure and provide trusted, familiar payment and notification workflows for local users. Together, these services significantly reduce development effort while ensuring Flex Work remains responsive, secure, and accessible to its target audience.

2.2.4 Quality Assurance and Testing

Quality assurance for Flex Work focuses on ensuring that all core functionalities operate smoothly, consistently, and without errors across different devices and usage scenarios. The application undergoes functional testing to verify that features such as user registration, job posting, job searching, application submission, and messaging perform as intended. UI and usability testing are also conducted to confirm that the interface is easy to navigate for non-technical users, with clear layouts and predictable interactions. Performance testing evaluates response times, real-time data updates, and the overall stability of the application under varying network conditions. In addition, Firebase Crashlytics is used to monitor crashes and errors, allowing developers to quickly identify and resolve issues. Through continuous testing and refinement, Flex Work maintains a reliable, user-friendly experience that meets the needs of both job seekers and employers.

2.2.5 Security and Privacy

Security and privacy are essential to Flex Work because the platform handles sensitive user information, job data, and financial transactions. To ensure safe access, the system uses Firebase Authentication, allowing users to register and log in through verified email or phone numbers. All communication between the mobile app and Firebase is secured through HTTPS encryption, while Firestore security rules restrict data access based on user roles, preventing unauthorized viewing or modification of personal information.

The application maintains confidentiality by implementing strict role-based permissions so that job seekers, employers, and admins can only access data relevant to their responsibilities. Messages, job details, applications, and reviews are stored securely in Firebase with controlled read/write privileges. For financial transactions, Flex Work integrates trusted local payment services such as Jazz Cash and EasyPaisa. These services use secure, tokenized payment flows, ensuring that sensitive account or wallet information is never stored inside the app or database.

To maintain overall system integrity, Firebase provides automated monitoring for suspicious activities, unusual access attempts, and potential security threats. Error tracking tools help identify issues that could compromise security or privacy. By combining encrypted communication, verified authentication, and controlled data access, Flex Work ensures a safe and trustworthy environment where users can confidently engage in job searching, hiring, and digital payments without fear of data misuse.

2.2.6 Scalability

Scalability is a core requirement of Flex Work, as the platform is designed to support a growing number of users, job postings, applications, and real-time interactions without compromising performance. To achieve this, the system uses Firebase's cloud-based infrastructure, which automatically adjusts to increased traffic, database load, and concurrent user activity. This allows the application to maintain fast response times even as more job seekers and employers join the platform.

The backend structure is built using Firestore collections designed to handle large data sets efficiently. Indexed queries ensure quick searching and filtering of job listings, while real-time listeners update user interfaces instantly when new jobs are posted or applications are submitted.

On the client side, Flex Work is optimized for a wide range of Android devices, ensuring consistent performance on both low-end and high-end smartphones. Lightweight UI components, efficient data-loading patterns, and background synchronization further support scalability. With its flexible architecture and cloud-driven backend, Flex Work is capable of expanding smoothly as user demand and system complexity grow over time.

2.2.7 Documentation

Documentation is a critical component of the Flex Work project, serving as the foundation for understanding, developing, testing, and maintaining the system throughout its lifecycle. It ensures that every phase of the project from requirement gathering to deployment is properly recorded and traceable. Comprehensive documentation also allows future developers, evaluators, or stakeholders to understand the system's structure, logic, and purpose without relying on verbal explanations. For a system like Flex Work, which integrates real-time communication, cloud backend services, and role-based interactions, proper documentation is essential for clarity, scalability, and long-term reliability.

The project documentation begins with the Software Requirements Specification (SRS), which defines the complete set of requirements necessary for the application's functionality. This includes the purpose of the system, its scope, user roles, functional modules, non-functional constraints, use case diagrams, and requirement identification techniques. The SRS outlines exactly what the system is expected to do and forms the contract between the development team and the intended system goals. It ensures that every feature such as registration, job

posting, messaging, payments, and admin control follows clearly defined rules and user expectations.

Following the SRS, the Software Design Description (SDD) provides a detailed breakdown of how these requirements are implemented technically. The SDD covers system architecture, process flows, sequence diagrams, class diagrams, data structures, and the logic behind major system components. It explains the interaction between the frontend Android application and the Firebase backend, including how data is stored, retrieved, validated, and displayed. The SDD also documents the design methodology used such as Agile and Iterative models which guided the order in which features were developed and refined. This ensures that the system is built in a structured, methodical, and maintainable way.

A complete set of UML diagrams is also included in the documentation to visually represent system behavior and interactions. These diagrams such as use case diagrams, sequence diagrams, class diagrams, and process flow diagrams help translate abstract ideas into visual models that are easy to understand. For example, use case diagrams clarify how job seekers, employers, and admins interact with the system; sequence diagrams illustrate the step-by-step execution of tasks like login, job posting, or payment processing; and class diagrams describe the relationships and responsibilities of different system entities. These diagrams play a key role in bridging the gap between requirements and implementation.

2.2.8 User Training and Support

User Training and Support are essential components of the Flex Work system, ensuring that all users whether job seekers, employers, or administrators can operate the application effectively and confidently. Because the platform is designed for a diverse audience, including individuals with limited digital literacy, the system must provide clear guidance, accessible instructions, and reliable support mechanisms. Proper training and support not only improve usability but also increase user trust, reduce errors, and encourage long-term adoption of the platform. For many users, Flex Work may be their first experience with a digital job marketplace, making intuitive guidance crucial.

The user training strategy begins with the design of the application itself. Flex Work incorporates a simple, clean interface that naturally guides users through different tasks without requiring formal training. Buttons, labels, and icons are deliberately chosen to be easily understood, even by users who are unfamiliar with technology. Key actions such as posting a job, applying for a job, or sending a message are presented clearly on the screen. Helpful text

prompts and validation messages appear when users make mistakes or skip required information, ensuring they can quickly correct errors without confusion. This built-in guidance reduces the need for external training materials and makes the system self-explanatory for first-time users.

To further support users, the application includes basic in-app explanations to help them understand the purpose of each feature and how to navigate the system. Simple help sections or tooltips clarify the meaning of icons, explain how to complete certain tasks, and offer quick troubleshooting steps for common problems such as login issues or failed submissions. Additionally, the platform encourages familiarity with processes like job posting or profile creation by using consistent layouts and predictable design patterns. The more consistent the experience, the easier it becomes for users to learn the system intuitively.

chapter 3

Requirement Analysis

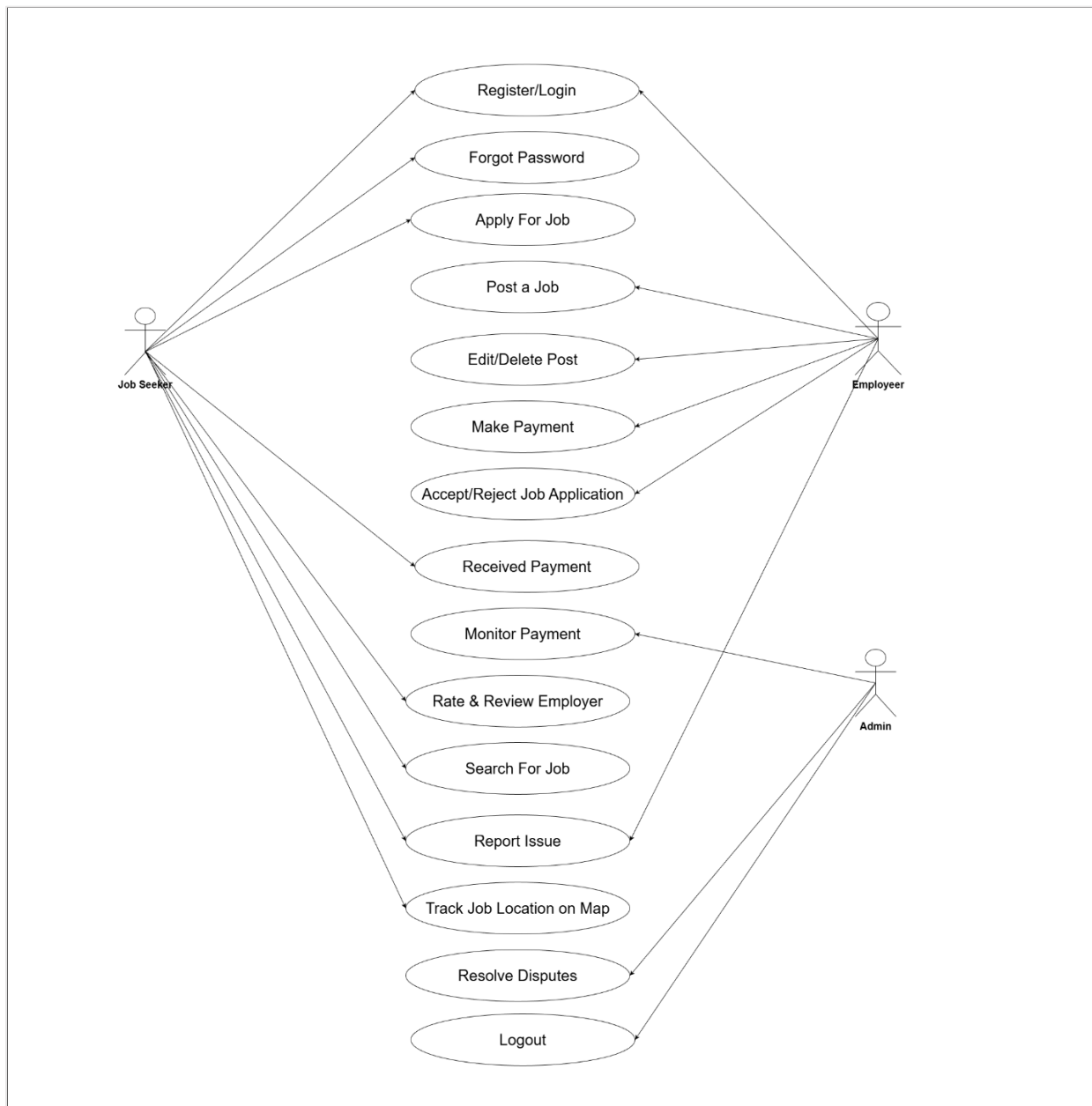
3 Requirement Analysis

Requirement Analysis is a fundamental phase of the software development lifecycle that ensures a clear understanding of what the system must achieve and how it should behave. This phase focuses on gathering, analysing, refining, and validating all the functional and non-functional requirements needed to design and build the Flex Work application. By identifying user needs early in the project, requirement analysis helps the development team minimize misunderstandings, reduce future modifications, and create a solution that directly aligns with the expectations of Job Seekers and Employers.

During this stage, discussions, observations, and system studies are used to translate user expectations into structured requirements. These requirements act as a blueprint that guides developers throughout the design, implementation, and testing phases. A well-conducted requirement analysis ensures that every feature such as job posting, profile management, messaging, notifications, and payment handling fits within the overall goals of the system and contributes to solving the defined problem.

Requirement analysis also establishes the boundaries and limitations of the system by determining what the application will do and what it will not do. It ensures clarity by differentiating between essential features and optional enhancements. In addition, this phase supports communication between stakeholders, ensuring that both technical teams and users share a mutual understanding of the system's functionality. Ultimately, requirement analysis sets the foundation for a successful, well-structured, and user-centered application that effectively meets the needs of the target audience.

3.1 Use Case Diagram



3.1.1 Use Case Diagram Explanation (Job Seeker, Employer & Admin)

The use case diagram for the Flex Work system illustrates the complete interaction between three primary actors: Job Seeker, Employer, and Admin. Each actor performs a distinct set of actions within the platform, and the diagram visually represents how these users engage with the core functionalities of the system. The purpose of this diagram is to provide a clear, high-level understanding of the system's scope and to define how responsibilities are distributed across different user roles. It also highlights shared actions, included functionalities, and system dependencies that support the application's workflow.

The Job Seeker interacts with several major use cases, beginning with essential account management features such as Register/Login and Forgot Password. Once authenticated, the Job Seeker can Search for Jobs, Apply for Jobs, Track Job Location on Map, and Report Issues related to the platform. They can also Receive Payments, Monitor Payments, and Rate & Review Employers after completing tasks. Many of these actions are connected through “include” relationships, indicating that processes like Apply for Job or Receive Payment require verification or secondary actions to complete successfully. These interactions reflect the Job Seeker’s journey throughout the system, from finding work opportunities to completing jobs and receiving digital payments.

The Employer actor has a parallel set of responsibilities that mirror the job posting and hiring process. Employers can Register/Login, Post a Job, Edit/Delete Posts, View Job Applications, Accept or Reject Job Seekers, and Make Payments for completed work. They also have access to rating features where they can Rate & Review Job Seekers. Employers interact with both the job management and payment modules extensively. The “include” connections in the diagram for example, between Make Payment and Receive Payment represent the shared interaction between Job Seeker and Employer during financial transactions. The interconnected nature of these use cases emphasizes the relational workflow between both parties.

The admin actor is responsible for platform oversight and system-level actions including Monitor Payment, Handle Reports, Resolve Disputes, and View Platform Statistics. Admins ensure that users follow the platform rules, disputes are resolved fairly, and fraudulent behavior is minimized. Their access allows them to monitor job activities, payment logs, and system performance. Admin use cases serve as the control mechanisms of the system and maintain platform integrity.

3.2 Use Case Descriptions

Table 3.2.1: Use Case Description of Register / Login

Use Case ID	UC-01
Use Case Name	Register / Login
Actor	Job Seeker, Employer, Admin
Description	This use case allows users to create an account or log in using email or phone number. The system verifies the user credentials and grants access to the platform according to their role.

Preconditions	User has a mobile device and stable internet connection.
Postconditions	User is successfully authenticated and redirected to their role-based dashboard.
Main Flow	1. User opens the app. 2. User selects Register or Login. 3. User enters required credentials. 4. System verifies the information. 5. User is granted access to their dashboard.

Table 3.2.2: Use Case Description of Search for Job

Use Case ID	UC-02
Use Case Name	Search for Job
Actor	Job Seeker
Description	The Job Seeker searches for available jobs using filters such as category, location, date, and pay range.
Preconditions	Job Seeker must be logged in.
Postconditions	Relevant job listings are displayed to the user.
Main Flow	1. Job Seeker opens the job search screen. 2. Applies filters (optional). 3. System retrieves matching jobs. 4. Results are displayed for browsing or applying.

Table 3.2.3: Use Case Description of Apply for Job

Use Case ID	UC-03
Use Case Name	Apply for Job
Actor	Job Seeker

Description	This use case enables the Job Seeker to submit an application for a selected job along with optional comments.
Preconditions	Job Seeker is logged in and has viewed job details.
Postconditions	Employer receives the application request.
Main Flow	1. Job Seeker selects a job. 2. Reviews job details. 3. Clicks “Apply”. 4. System records the application. 5. Employer is notified.

Table 3.2.4: Use Case Description for Post

Use Case ID	UC-04
Use Case Name	Post a Job
Actor	Employer
Description	The Employer posts a new job by entering details such as title, description, pay, location, and time.
Preconditions	Employer must be logged in.
Postconditions	Job listing becomes visible to Job Seekers.
Main Flow	1. Employer opens “Post Job”. 2. Enters job information. 3. Confirms submission. 4. System validates and saves the job. 5. Job is published in the listings.

Table 3.2.5: Use Case Description for Accept / Reject Application

Use Case ID	UC-05
Use Case Name	Accept/Reject Application
Actor	Employer

Description	Employer reviews applications submitted by Job Seekers and decides whether to accept or reject them.
Preconditions	Employer has a posted job with received applications.
Postconditions	Job Seeker is notified of the decision.
Main Flow	1. Employer views all applications. 2. Selects a specific applicant. 3. Chooses Accept or Reject. 4. System updates application status. 5. Job Seeker receives a notification.

Table 3.2.6: Use Case Description for Make Payment

Use Case ID	UC-06
Use Case Name	Make Payment
Actor	Employer
Description	Employer makes payment to the Job Seeker via integrated Jazz Cash or EasyPaisa services after job completion.
Preconditions	Job must be completed and approved.
Postconditions	Payment is transferred and recorded in transaction history.
Main Flow	1. Employer selects job. 2. Opens payment screen. 3. Selects Jazz Cash/EasyPaisa. 4. Confirms amount. 5. System processes and updates payment status.

Table 3.2.7: Use Case Description for Receive Payment

Use Case ID	UC-07
Use Case Name	Receive Payment
Actor	Job Seeker

Description	Job Seeker receives digital payment once the Employer completes the transaction.
Preconditions	Employer has initiated payment.
Postconditions	Job Seeker receives confirmation and balance update.
Main Flow	1. Employer sends payment. 2. System validates transaction. 3. Job Seeker receives confirmation. 4. Payment is added to their account/wallet.

Table 3.2.8: Use Case Description for Report Issue

Use Case ID	UC-08
Use Case Name	Report Issue
Actor	Job Seeker, Employer
Description	Users can report issues related to jobs, payments, or user behaviour.
Preconditions	User must be logged in.
Postconditions	Admin receives the report.
Main Flow	1. User selects “Report Issue”. 2. Describes the problem. 3. System logs the report. 4. Admin reviews the issue.

Table 3.2.9: Use Case Description for Resolve Disputes

Use Case ID	UC-09
Use Case Name	Resolve Disputes
Actor	Admin

Description	Admin resolves conflicts between Job Seekers and Employers regarding jobs, payments, or misconduct.
Preconditions	A dispute has been submitted.
Postconditions	Dispute is resolved and users are notified.
Main Flow	1. Admin views dispute. 2. Investigates details. 3. Reviews evidence provided. 4. Takes action. 5. Updates status and notifies users.

Table 3.2.10: Use Case Description for Monitor Platform Statics

Use Case ID	UC-10
Use Case Name	Monitor Platform Statistics
Actor	Admin
Description	Admin monitors jobs, applications, user activity, and payment statistics.
Preconditions	Admin is logged in.
Postconditions	Updated platform metrics are displayed.
Main Flow	1. Admin opens dashboard 2. Views statistics. 3. System displays graphs and logs. 4. Admin identifies trends or issues.

3.3 Functional Requirements

The functional requirements of the Flex Work system define the essential operations, behaviours, and interactions that the application must support in order to fulfill its intended purpose. Since the platform is designed to connect Job Seekers and Employers while being monitored by an Admin, the system must provide clear, reliable, and interactive features that accommodate each role. These requirements describe the expected system responses, data processes, and user actions, ensuring that all functionalities work together cohesively to support real-time job searching, hiring, communication, and payments.

The system must allow users to register and log in using verified credentials such as email or phone number, ensuring that each user has a unique identity associated with their role. Once authenticated, users should be able to manage their profiles, update personal details, and maintain accurate information for job searching or hiring decisions. The platform must enable Employers to post jobs by entering essential details like description, pay rate, job location, and timing. They should also be able to edit or delete these job posts whenever necessary. On the other hand, Job Seekers must be able to browse, search, and filter available jobs according to categories, location, date, or pay, and then apply directly through the app.

Core interactions such as job applications must trigger system responses where Employers can view applicants and decide whether to accept or reject them. These actions must automatically notify the Job Seeker about their application status. The system should also support in-app messaging, allowing both parties to communicate easily regarding job details, availability, or clarifications. After job completion, the Employer must be able to make payments using integrated Jazz Cash or EasyPaisa services, while Job Seekers must receive these payments securely and view their transaction history. Additionally, both users should have an option to provide ratings and reviews, enabling a feedback system that helps build trust and credibility on the platform.

To maintain fairness and platform integrity, the system must allow users to report issues or disputes related to jobs, payments, or user behaviour. These reports should be forwarded to the admin, who can monitor platform activity, investigate complaints, and take necessary actions such as resolving disputes or moderating content. Admins must also have access to system statistics, payment logs, and user activities to ensure smooth functioning of the application. The system must support real-time notifications for essential events such as new jobs posted, application updates, messages received, payments processed, and admin actions. Finally, every

user must be able to log out securely from the system, ensuring that their session is closed and their data remains protected.

These functional requirements collectively ensure that Flex Work operates as a complete job-matching ecosystem, supporting all necessary interactions from registration to job completion, payments, communication, and dispute resolution. They form the foundation for the system's design, ensuring reliability, clarity, and consistency in user experience. [3]

3.3.1 User Registration and Login

User Management is one of the most fundamental functional areas of the Flex Work system, as it ensures that every individual accessing the platform can be identified, authenticated, and managed according to their role. The User Registration and Login functionality provides a secure gateway for Job Seekers, Employers, and Admins to enter the system and interact with its features. This module ensures that only verified and authorized users gain access to the platform, helping maintain data integrity, system security, and reliable communication among participants.

The registration process allows new users to create an account by providing essential details such as name, email address or phone number, and password. The system validates the information and verifies the account through secure methods such as email or SMS confirmation. This verification step ensures that fake or untraceable accounts do not enter the system, protecting both Job Seekers and Employers from fraudulent interactions. Upon successful registration, the user is assigned a role either Job Seeker or Employer which determines the features and permissions available to them within the application.

The login process enables returning users to access their accounts using their registered credentials. The system checks the provided information against stored records in Firebase Authentication and grants access only if the credentials are valid. The login module also ensures that user sessions are securely managed, preventing unauthorized access to personal or financial data. If a user enters incorrect credentials, the system displays an appropriate error message, guiding them to correct the mistake without compromising system security. Additionally, a “Forgot Password” option is provided to help users recover access if they lose their login information, ensuring that account recovery is quick, secure, and user-friendly.

Overall, User Registration and Login form the entry point to the entire Flex Work ecosystem. By ensuring secure authentication, role-based access, and verified identities, this functionality

establishes a trustworthy environment where Job Seekers, Employers, and Admins can interact confidently and responsibly. This module supports the system's goal of providing a reliable digital marketplace where all users are authenticated and their actions are traceable, contributing significantly to platform safety and operational stability.

3.3.2 Profile Management

Profile Management is an essential functional component of the Flex Work system, enabling users to maintain accurate and up-to-date personal information that supports effective job matching and communication. This module ensures that both Job Seekers and Employers can manage their identity within the platform, providing transparency and trust between parties. A well-maintained profile helps Employers evaluate applicants more accurately and allows Job Seekers to present their skills, experience, and work history clearly. For Employers, their profile reflects business details, job posting history, and credibility on the platform. As such, Profile Management directly influences user engagement, job acceptance decisions, and overall platform reliability.

The Profile Management process allows users to view and edit their personal details at any time. Job Seekers can update information such as name, location, age, contact number, skill set, daily availability, and past work experience. This ensures that job applications reflect the most accurate representation of their capabilities. Employers can similarly update business information, including company name, address, job categories they frequently post, and preferred communication details. The system validates all updates to ensure they meet required formats and that user data remains consistent and structured across the platform. Any changes made to the profile are immediately reflected in Firebase, ensuring real-time synchronization across all devices.

3.3.3 Initialization and Main Menu

The Initialization and Main Menu component serves as the starting point for user interaction within the Flex Work application. Once the system is launched, it performs essential initialization tasks such as loading necessary resources, establishing a secure connection with Firebase services, and checking the user's authentication status. If the user is already logged in, the system automatically retrieves their role Job Seeker, Employer, or Admin and redirects them to the appropriate dashboard without requiring additional input. This reduces login fatigue, improves responsiveness, and ensures a seamless user experience. If no active session

exists, the system displays the initial screens prompting the user to either log in or register, guiding them smoothly toward accessing the platform.

After initialization, the Main Menu serves as the central navigation hub that allows users to access the primary features of the application. The menu layout is customized based on the user's role, ensuring that each user type only sees options relevant to their permitted actions. For Job Seekers, the Main Menu typically includes options such as searching for jobs, viewing job applications, accessing the map for job locations, managing their profile, checking notifications, sending messages, and viewing payments received. Employers, on the other hand, have access to options like posting new jobs, managing existing job listings, reviewing applicants, communicating with workers, processing payments, and managing profile details. This role-based menu structure ensures clarity, prevents confusion, and supports faster task completion.

The Main Menu is designed to be simple, visually clean, and intuitive so that users with varying levels of technical literacy can navigate the system without difficulty. Icons and labels are chosen to make menu options self-explanatory, while the consistent layout across screens creates familiarity and ease of use. The system also displays real-time indicators such as new notifications or pending actions, helping users prioritize important tasks. Through Initialization and Main Menu functions, the Flex Work system establishes a reliable entry point and an organized navigation structure that supports efficient user interaction and creates a foundation for the entire application workflow.

3.3.4 Inventory Management (POS)

The Inventory Management (POS) module is designed to facilitate employers who manage physical products, tools, or job-related materials within the Flex Work ecosystem. This feature is particularly useful for small businesses, workshop owners, or service providers who rely on an internal stock of items such as cleaning supplies, construction tools, repair equipment, or delivery materials. The POS-based inventory system helps employers maintain accurate records of items used during job assignments, track stock availability, and ensure that essential materials are always prepared before work is assigned to a Job Seeker. By integrating a basic POS workflow, the system supports operational efficiency and improves employer readiness for upcoming jobs.

The Inventory Management module allows employers to add new items into their inventory by specifying details such as item name, quantity, cost, and stock category. Whenever a job

requires certain tools or materials, the employer can link those items to the job post, enabling the system to automatically deduct the required quantity once the job is marked as completed. This ensures accurate consumption tracking and prevents stock discrepancies. Additionally, employers can update item quantities, restock items, and remove outdated products from their inventory. The system maintains a real-time record of stock levels using Firebase database synchronization, ensuring that all changes appear instantly across devices.

A significant feature of this module is its ability to generate simple POS-style summaries that help employers monitor material usage, cost allocation, and stock performance over time. These summaries provide insight into which items are frequently used, which require restocking, and how much operational cost is associated with different job categories. Alerts can also be generated when stock reaches a low threshold, prompting employers to replenish essential items. Through these combined functionalities, the Inventory Management (POS) module enhances transparency, improves resource management, and ensures that employers can operate more efficiently within the Flex Work platform.

3.3.5 Update Notifications

The Update Notifications functionality is a critical component of the Flex Work system, ensuring that users are informed in real time about important events and changes related to their activities. Since the platform connects Job Seekers and Employers in a dynamic environment where jobs are posted, applications are submitted, payments are processed, and issues are resolved continuously, timely communication is essential. The notification system is designed to keep users engaged and up to date without requiring them to constantly check the application manually. This improves responsiveness, reduces delays in decision-making, and enhances the overall usability of the system.

Whenever an action occurs that affects a user such as a new job being posted in their area, an application being accepted or rejected, a new message being received, a payment being processed, or a dispute being updated the system generates a notification and delivers it to the relevant user. These notifications are implemented using Firebase Cloud Messaging (FCM), which allows the app to send push notifications even when it is running in the background or not actively open. For Job Seekers, notifications may include alerts for new jobs matching their preferences, changes in application status, payment confirmations, or messages from Employers. For Employers, notifications may indicate new applications on posted jobs, worker

responses, job completion confirmations, or payment-related updates. Admins can also receive alerts for reported issues or system events requiring review.

The design of Update Notifications ensures that they are clear, concise, and informative, displaying essential information such as job title, action type, and the party involved. Tapping a notification directs the user to the relevant screen in the application, allowing them to respond quickly for example, reviewing an application, replying to a message, or confirming a payment. This functionality not only improves the efficiency of user interactions but also supports continuous engagement with the platform. By providing real-time updates in a structured and meaningful way, the Update Notifications module significantly enhances communication, reduces uncertainty, and strengthens the reliability of the Flex Work ecosystem. [4]

3.3.6 Product Description and Customization

The Product Description and Customization module allows Employers to provide detailed information about the services or job requirements they are offering and, when applicable, customize certain job attributes to fit their specific needs. Although Flex Work primarily focuses on job-based interactions rather than physical products, many employers especially small businesses, service providers, and workshop owners require a flexible way to define the nature of the job, associated tasks, required skills, and additional conditions that may vary from one assignment to another. This functionality ensures that every job listing is clear, structured, and tailored, helping Job Seekers make informed decisions before applying.

3.3.7 General Features

The General Features module represents the collection of broad, system-wide capabilities that enhance the overall usability, accessibility, and performance of the Flex Work application. These features support the core functionality of the system and ensure that users experience a smooth, intuitive, and secure interaction across all modules. General features are not tied to a specific role or task; instead, they contribute to the platform's stability, consistency, and user satisfaction. They include essential elements such as navigation design, responsive layout, multilingual support, real-time updates, notifications, and error-handling mechanisms.

The application provides a clean and intuitive interface that allows users to access key functions with minimal effort. A structured navigation system ensures that Job Seekers, Employers, and Admins can easily move between sections such as job listings, messages, profiles, payments, and reports. The layout is responsive across a wide range of Android devices, ensuring that screens adjust smoothly to different sizes and resolutions. The system also incorporates

consistent design patterns such as standardized buttons, icons, menus, and colour schemes to enhance familiarity and reduce the learning curve, especially for users who may not be technologically experienced.

Another key general feature is the system's real-time responsiveness. Flex Work uses Firebase to ensure that updates such as new job posts, application changes, payments, and notifications are reflected instantly without requiring users to refresh the application manually. This greatly improves communication efficiency and supports time-sensitive interactions. In addition, the system includes multilingual support and easily understandable labels, making the platform more accessible to users across different backgrounds. Error-handling and validation mechanisms are also integrated to guide users when incorrect or incomplete inputs are detected, ensuring that all processes are performed smoothly and correctly.

Overall, the General Features module ensures that the Flex Work platform remains reliable, user-friendly, and adaptable. By combining intuitive design, responsive behaviour, real-time data synchronization, and accessible user guidance, these features strengthen the overall quality and user experience of the system, supporting its broader mission of enabling seamless digital job matching for diverse user groups.

3.3.8 Real-Time Notifications

Real-Time Notifications are a crucial feature of the Flex Work system, enabling immediate communication between Job Seekers, Employers, and Admins. Because the platform operates in a dynamic environment where job posts, applications, payments, and messages are constantly updated, the system must notify users instantly about events relevant to their role. Without a real-time mechanism, users may miss job opportunities, application responses, or important updates regarding disputes or payments. This feature ensures that Flex Work remains responsive, interactive, and engaging, significantly improving communication speed and user satisfaction across the platform.

The system uses Firebase Cloud Messaging (FCM) to deliver push notifications directly to users' devices, even when the app is running in the background. When key events occur such as an Employer posting a new job, a Job Seeker submitting an application, an Employer accepting or rejecting an applicant, or a payment being processed the system immediately triggers a notification. These notifications contain concise, meaningful information, allowing users to quickly understand the action and respond accordingly. For Job Seekers, notifications may include alerts for new jobs matching their preferences, application decisions, or updates

in payment status. Employers receive alerts when new applications arrive, when Job Seekers respond, or when disputes require attention. Admins may receive notifications about reported issues or critical system events that require review.

The real-time communication enabled by this feature supports efficient workflows and reduces uncertainty among users. By tapping on a notification, the user is redirected to the appropriate screen inside the application, ensuring that important tasks can be completed without delay. This feature also enhances the overall reliability of the system by keeping users continuously informed of changes that directly affect their interactions and responsibilities. Through instant updates and seamless integration with backend events, the Real-Time Notifications module strengthens user engagement, facilitates fast decision-making, and contributes to a transparent and interactive experience within Flex Work. [5]

3.3.9 Training and Tutorial System

The Training and Tutorial System is an essential support feature within Flex Work, designed to help users understand how to navigate the application and utilize its key functionalities effectively. Because the platform serves a diverse audience including users who may have limited digital literacy it is important that the system provides accessible and easy-to-understand guidance. This feature ensures that Job Seekers and Employers can confidently interact with the application without external assistance, reducing user frustration and increasing overall system adoption. By providing built-in learning resources, the Training and Tutorial System enhances usability and ensures that important tasks such as job posting, applying for jobs, messaging, and payments are understood clearly by all users.

The system includes a series of simple, guided tutorials that appear during first-time use or can be accessed anytime through the help menu. These tutorials explain the purpose of major screens, demonstrate how to perform core tasks, and highlight essential buttons or icons. The tutorials are kept short, visual, and easy to follow, consisting of pop-up instructions, arrows pointing to key features, and short descriptions that guide users step-by-step. For example, new Job Seekers are shown how to search for jobs, apply, view their application status, and communicate with Employers. Employers receive guidance on posting a job, viewing applicants, responding to messages, and making payments. Admins can also access tutorial instructions related to monitoring tools and dispute resolution.

In addition to guided walkthroughs, the Training and Tutorial System provides a help section that includes frequently asked questions, visual illustrations, and short explanations of common

issues users may face. This ensures that support is available even after the initial onboarding process. The tutorials are designed to be accessible on all device sizes and require minimal reading, making them suitable for users with varying levels of education and technical experience. Overall, the Training and Tutorial System strengthens the reliability, accessibility, and user-friendliness of the Flex Work platform by empowering users with the knowledge they need to successfully navigate the system and carry out their tasks with confidence.

3.4 Non-Functional Requirements

Non-Functional Requirements define the quality attributes, performance expectations, and operational constraints of the Flex Work system. These requirements ensure that the application is not only functional but also secure, reliable, efficient, and user-friendly. They describe how the system should behave and the standards it must meet across various technical and usability dimensions.

3.4.1 Performance Requirements

The Flex Work system must operate efficiently under normal and high-usage conditions. The application should load primary screens such as the job listings, dashboard, and messages within a few seconds, even when the database contains large volumes of job posts or user records. Real-time updates such as notifications, application status changes, and messaging events should appear instantly, ensuring smooth interaction between Job Seekers and Employers. The system must handle multiple concurrent operations without delays, including simultaneous job postings, user logins, message exchanges, and payment processing. The Firebase backend should support dynamic scaling to accommodate an increasing number of users as the platform grows, ensuring consistent performance at all times.

3.4.2 Usability Requirements

Flex Work must be simple, intuitive, and easily usable by individuals with different levels of technical literacy. The interface should be clean, consistent, and easy to navigate, with clear labels, icons, and instructions guiding users through each task. All essential features should be accessible within a few taps, minimizing the cognitive load for users who are not familiar with mobile applications. Error messages should be descriptive and helpful, allowing users to quickly correct mistakes. Fonts, colours, and layout spacing must be chosen carefully to ensure readability across all Android devices. The system must also support a quick learning curve, making it easy for first-time users to navigate the app without external assistance.

3.4.3 Reliability Requirements

The system must perform consistently without unexpected failures, data loss, or operational interruptions. Firebase must ensure persistent data storage so that job posts, applications, and user profiles remain intact even after app restarts or network changes. In the case of network disruption, the application should handle the situation gracefully by displaying offline messages or retrying operations when connectivity is restored. The system should also maintain data integrity by ensuring that duplicate entries, incomplete submissions, or partial transactions do not occur. Scheduled maintenance processes and automated backups must preserve critical data and ensure quick recovery if any unexpected issue arises.

3.4.4 Security Requirements

Security is a critical concern due to the sensitive personal and payment information stored within the system. The application must provide secure authentication through Firebase Authentication to prevent unauthorized access. All data transmission between the mobile application and backend services should occur over encrypted HTTPS channels to protect against interception or manipulation. Access to Firestore collections must be governed by strict security rules, ensuring that users only view and modify data relevant to their role. Payment transactions handled through Jazz Cash and EasyPaisa must follow secure tokenized processes, ensuring no sensitive financial data is stored within the app. The system should also log suspicious activities and enforce automatic logout after extended inactivity.

3.4.5 Scalability Requirements

Flex Work must be designed to grow without requiring major architectural changes. The Firebase backend should support increasing numbers of users, job posts, messages, and transaction records without affecting speed or stability. The database structure must be optimized using indexes to ensure fast queries, even when handling thousands of entries. The codebase must follow a modular architecture, enabling new features such as advanced filtering, AI-based recommendations, or employer analytics to be added easily. The system must support varying device types, from low-end to high-end phones, ensuring smooth performance across all supported Android versions.

3.4.6 Maintainability Requirements

The system must be easy to update, debug, and extend over time. The code should follow clean, modular, and well-documented practices, enabling developers to identify issues and make improvements without affecting unrelated modules. All backend rules, frontend screens, and

logical workflows should be structured clearly to support long-term maintenance. Firebase Crashlytics and logging tools should provide detailed error reports, enabling quick diagnosis of bugs. The documentation including SRS, SDD, UML diagrams, and configuration files must be kept updated to ensure future developers can understand system components easily.

Chapter 4

Design and Architecture

4 Design and Architecture

4.1 System Overall Architecture

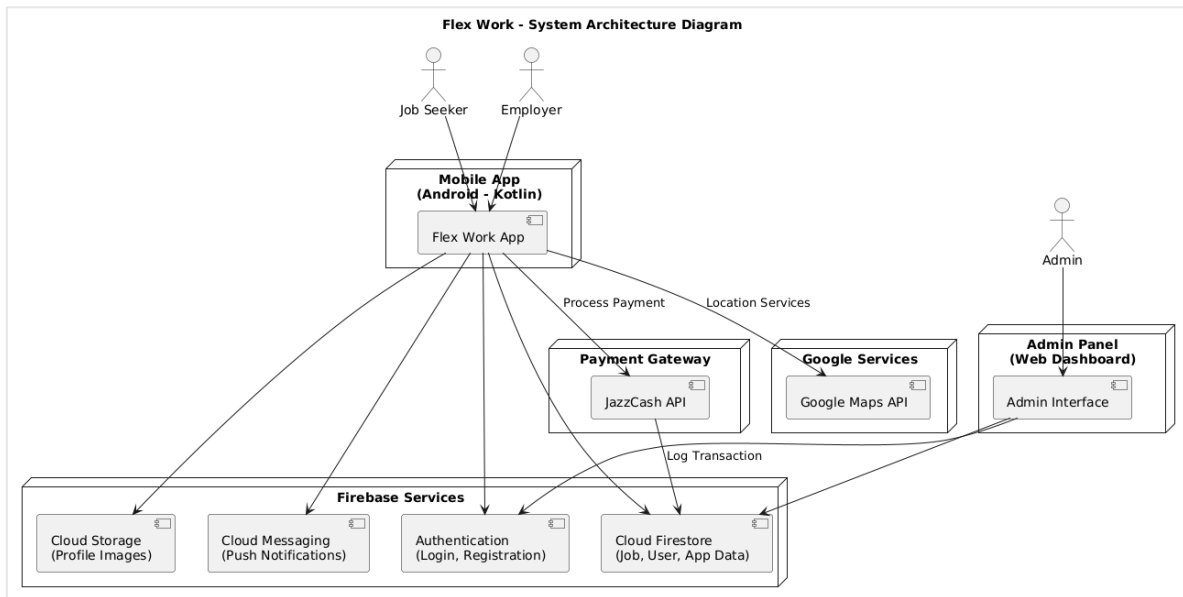


Figure 4.1 Architecture Diagram

The system architecture of Flex Work follows a modular, scalable, and service-oriented design that supports real-time communication, secure authentication, and smooth data handling across different user roles. The architecture is built on a client-server model where the Android mobile application functions as the client interface, and Firebase provides the backend infrastructure. This approach ensures high availability, low latency, and seamless synchronization of data across all devices. The architecture is designed to support the three major user roles Job Seeker, Employer, and Admin each interacting with the application through defined workflows and permissions.

At the core of the architecture lies Firebase Authentication, Firebase Storage, Firebase Cloud Messaging, and Firebase Storage, all of which work together to deliver fast, secure, and reliable operations. Firebase Authentication manages user identity, ensuring that only legitimate users access the system. Firebase Storage serves as the primary database, storing structured data such as job listings, user profiles, applications, messages, and payment records. Its real-time synchronization capabilities allow instant updates across the application, enabling features such as live chat, real-time notifications, and instant updates on job applications. Firebase Storage manages media files like profile images and uploaded documents, ensuring efficient and secure file handling.

4.1.1 Client-Side Mobile Application

The client side consists of the Android mobile application developed using Kotlin. This layer includes all user interface elements, user interaction handlers, and presentation logic. The application displays dashboards, job listings, application forms, chat screens, notifications, and payment interfaces based on each user's role. It collects user inputs and forwards them to backend services through secure API calls. The client is responsible for rendering smooth UI transitions, validating local inputs, and providing responsive feedback to ensure a pleasant user experience. It also integrates Firebase SDKs to communicate with backend services in real time.

4.1.2 Firebase Authentication Layer

The Authentication layer manages user identities, credentials, and secure login sessions. Firebase Authentication is used to register and authenticate Job Seekers, Employers, and Admins using email or phone numbers. This layer ensures that only verified users can access the system and that each user is identified with a unique UID. Authentication tokens are managed by Firebase, reducing security risks and simplifying the login process. This layer also supports password recovery and session management. [6]

4.1.3 Firebase Storage Database Layer

Firebase Storage serves as the primary data storage layer of Flex Work. It is used to store job posts, user profiles, applications, messages, payment records, and reports. The database is structured using collections such as Users, Jobs, Applications, Messages, and Reports. Firestore's real-time synchronization ensures that all users instantly receive updates such as new jobs, application changes, or incoming messages. This layer is optimized for high-performance queries using indexed fields. Firestore Security Rules enforce role-based access control, ensuring that Job Seekers, Employers, and Admins can only access data permitted for their roles.

4.1.4 Cloud Storage Layer

Firebase Cloud Storage is used to store user-uploaded files such as profile pictures, attachments, or job-related images. This layer provides secure, scalable, and fast content delivery. The client application accesses Cloud Storage through secure download URLs managed by Firebase. Access is controlled using Firebase rules to ensure files remain protected and only authorized users can view or modify them.

4.1.5 Real-Time Messaging & Notification Layer

This layer supports real-time interactions across the system. Firebase Cloud Messaging (FCM) enables push notifications for job updates, application decisions, messages, and payments. The messaging module lets Job Seekers and Employers communicate instantly through real-time Firestore listeners. Every interaction such as new messages or application updates is immediately reflected on all connected devices without the need for manual refresh. This ensures efficient communication and enhances user responsiveness.

4.1.6 Payment Integration Layer (Jazz Cash & EasyPaisha)

Flex Work integrates local payment gateways Jazz Cash and EasyPaisha to facilitate digital payments between Employers and Job Seekers. This layer communicates with the payment APIs to process transactions securely. It ensures that payment requests, confirmations, and transaction logs are recorded in Firestore. The architecture isolates payment functions within a dedicated module to ensure security and ease of maintenance.

4.1.7 Admin Dashboard & Monitoring Layer

This layer handles administrative operations including user management, job monitoring, report handling, and dispute resolution. Admins interact with a specialized interface that retrieves system-wide data from Firestore. The dashboard displays statistics such as active jobs, transactions, reported issues, and user behaviour logs. The admin layer ensures system stability, detects misuse, and enforces community guidelines.

4.1.8 System Integration and Workflow Coordination

This layer ties together all modules client, backend services, messaging, and payments into a cohesive workflow. The integration layer ensures seamless transitions among tasks such as applying for jobs, messaging, and completing payments. It handles data flow between components and ensures consistency across the system. The architecture uses Firebase's real-time listeners to coordinate workflows, making the platform intuitive and responsive.

4.1.9 Scalability and Extensibility Considerations

The architecture is designed for future expansion. Modular code structure allows additional features like advanced analytics, or worker verification to be added easily without disrupting existing modules. Firebase's serverless infrastructure automatically scales with traffic, eliminating the need for manual server management.

4.2 Process Flow/Representation

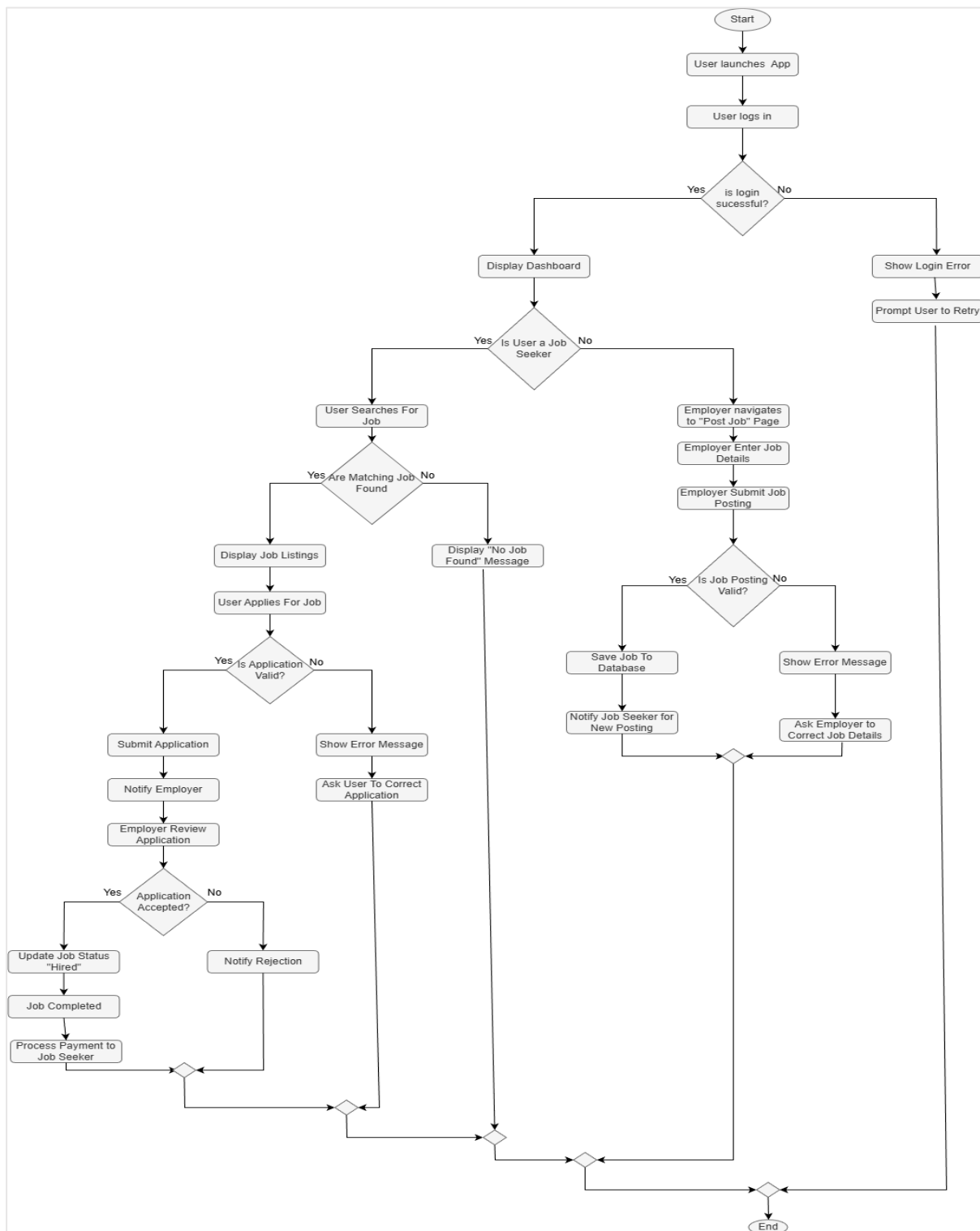


Figure 4.2 Process Flow Diagram

The System Flow of Flex Work describes how users interact with the platform and how data moves between different components of the system. It provides a high-level view of the logical sequence of operations, demonstrating how the application transitions from one step to another

to complete essential tasks such as registration, job searching, job posting, payments, messaging, and dispute handling. The flow ensures that all modules work together cohesively, following a structured pattern that maintains consistency and reliability across the system.

The system flow begins with the initialization phase, where the application checks for an active user session. If the user is already authenticated, they are redirected to their respective dashboard based on their assigned role Job Seeker, Employer, or Admin. New users are guided to the registration or login process, where their credentials are verified through Firebase Authentication. Once verified, the system retrieves their profile information from Firebase Storage and loads the appropriate interface. This structured onboarding process ensures secure access and smooth entry into the system for all user types.

For Job Seekers, the system flow typically follows these steps: viewing the dashboard → searching for available jobs → exploring job details → submitting an application → receiving Employer responses → messaging for clarifications → completing the job → receiving payment → submitting a rating or review. Each action triggers associated backend processes, such as updating application status, sending real-time notifications, or logging payment transactions. The flow ensures that Job Seekers remain informed and can navigate their tasks efficiently.

For Employers, the system flow begins with accessing their dashboard → posting a job → viewing incoming applications → selecting or rejecting Job Seekers → communicating through the messaging module → marking job completion → sending payment via Jazz Cash or EasyPaisa → reviewing the worker's performance. The flow ensures that Employers maintain full control over job management, applicant filtering, and communication channels, allowing for smooth coordination with Job Seekers.

4.3 Sequence Diagram

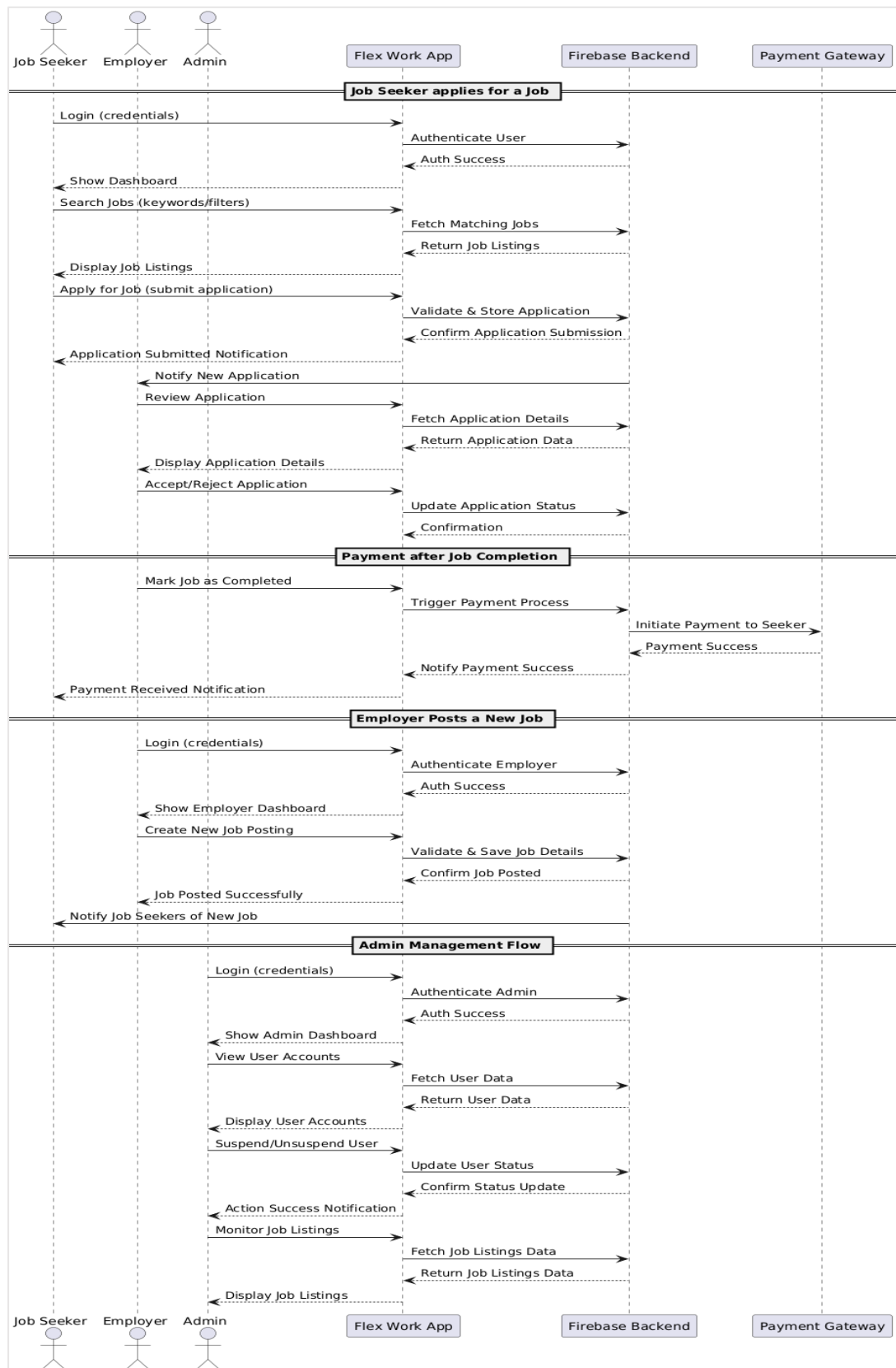


Figure Error! Use the Home tab to apply 0 to the text that you want to appear here..1.3 Sequence Diagram

The sequence diagrams in Flex Work illustrate the step-by-step interaction between users, the mobile application interface, and the Firebase backend components. These diagrams provide a clear picture of how data flows through the system during various processes, revealing the order of operations and the communication between different modules. By modelling the timing and interaction sequence, these diagrams help developers understand how the application responds to user actions, how backend services are invoked, and how real-time updates are delivered. Sequence diagrams also assist in verifying logical correctness and ensuring that workflows follow the intended behaviour defined in the requirements.

One of the primary sequence diagrams represents the User Registration and Login process. The sequence begins when a new user opens the application and selects the registration option. The system prompts the user to enter essential information such as name, email or phone number, and password. Once submitted, the application sends the data to Firebase Authentication, which validates the input and creates a new user account if the information is correct and not already registered. After successful registration, the system stores the user's role and profile details inside Firebase Storage. The system then returns a success response, and the user is redirected to their role-based dashboard. For returning users, the login sequence involves entering credentials, sending them to Firebase Authentication for verification, receiving validation, retrieving the user profile from Firestore, and loading the appropriate interface.

The sequence diagram ensures that all interactions in this process are structured and secure. It highlights the verification steps, user feedback, backend communication, and data retrieval from Firestore. Additionally, the diagram clarifies how the application handles incorrect input or failed authentication through error messages and retry prompts. The process demonstrates real-time synchronization between the mobile client and backend services, ensuring that user sessions are properly initialized and authorized. By defining these interactions clearly, the sequence diagram helps maintain a secure, streamlined, and user-friendly authentication workflow within the Flex Work platform.

4.4 Class Diagram:

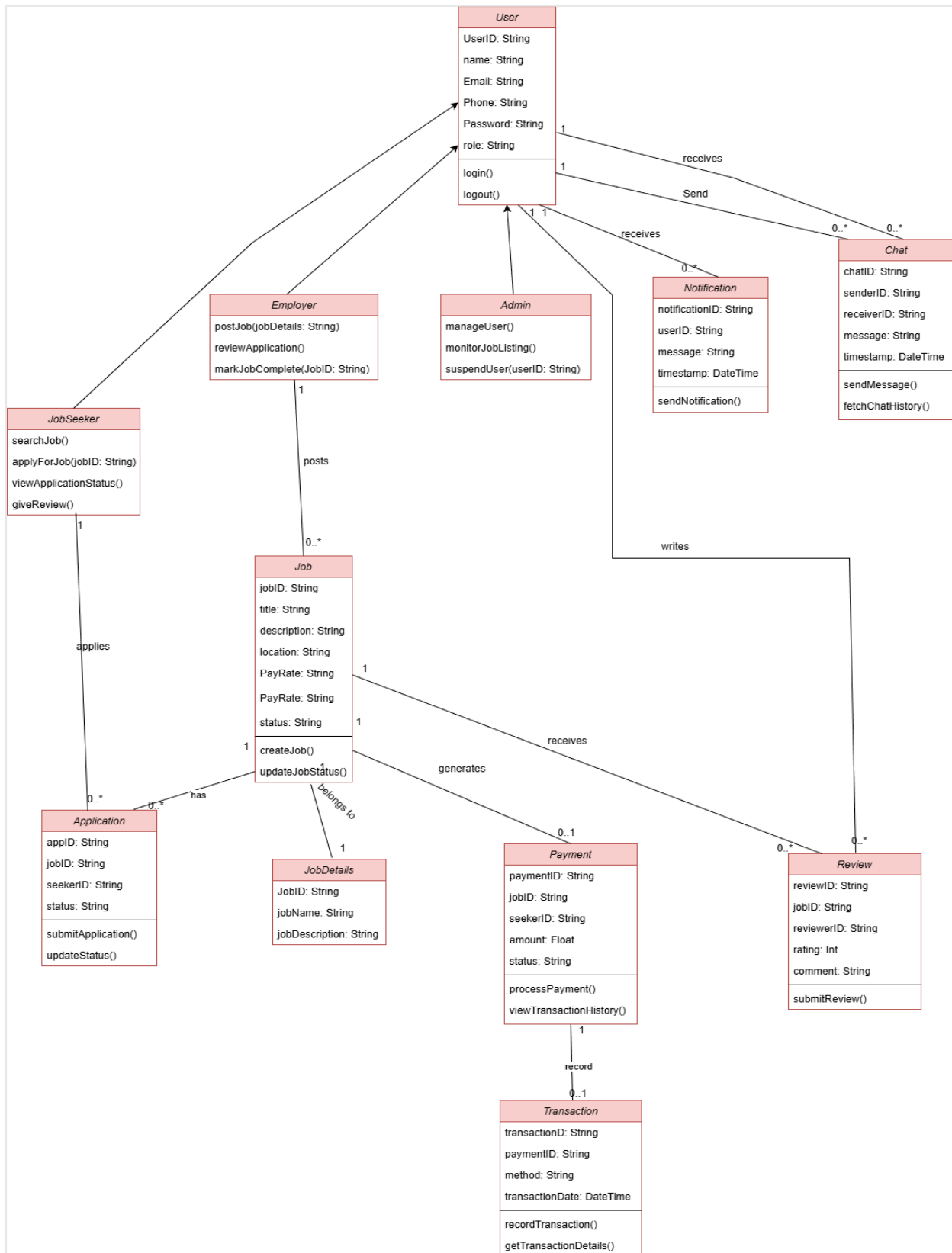


Figure 4.4 Class Diagram

The Class Diagram of Flex Work represents the structural foundation of the system by illustrating the major classes, their attributes, operations, and the relationships among them. It serves as a blueprint for developers and provides a static view of the system's object-oriented design. By organizing the system into logical classes such as User, Job, Application, Payment, Message, and Report, the diagram ensures clarity, maintainability, and scalability throughout the development process. Each class encapsulates its own data and behaviours, supporting a modular structure that aligns with the core features of the Flex Work platform.

At the centre of the diagram is the User class, representing all types of users including Job Seekers, Employers, and Admins. The User class includes attributes such as user ID, name, email or phone number, role, and profile details. It provides methods for authentication, updating profile information, and managing account settings. Two specialized roles Job Seeker and Employer extend this base class through inheritance, gaining additional attributes and behaviours based on their specific responsibilities. Jobseeker contains information such as skills, experience, and completed jobs, while Employer contains business information, job posting history, and payment-related details. The admin role is also derived from the User class and includes functions related to system monitoring and dispute handling. [7]

4.4.1 User-Related Classes

- **User Class (Base Class)**

The User class serves as the parent class for all system users. It contains common attributes such as userID, name, email/phone, password, role, and profile Image. It also includes operations for authentication, updating profile information, and managing account settings. This abstraction provides a reusable framework that reduces redundancy across user types.

- **JobSeeker Class (Derived Class)**

This class inherits from User and adds Job Seeker specific attributes such as skills, experience, location, and job history. It provides methods for searching jobs, applying for jobs, receiving payments, and reviewing Employers.

- **Employer Class (Derived Class)**

Employer also inherits from *User* but includes additional attributes like business Name, business Type, posted job history, and payment responsibilities. Operations include posting jobs, managing applicants, communicating with Job Seekers, and sending payments.

- **Admin Class (Derived Class)**

The admin class extends from User and performs system level operations such as monitoring user activity, handling disputes, reviewing reports, and enforcing system policies.

4.4.2 Job and Application Classes

- **Job Class**

The Job class represents all job postings created by Employers. Key attributes include jobID, title, description, location, payrate, category, and employed. It is directly linked to the Employer who created it. Methods include creating, editing, deleting, and publishing job details.

- **Application Class**

This class links Job Seekers to job opportunities. It stores application ID, job ID, seeker ID, status, and application timestamps. Employers interact with this class to review, accept, or reject applications. Multiple applications may be associated with a single job, represented by a one-to-many relationship.

4.4.3 Communication Classes

- **Message Class**

The Message class facilitates real-time communication between Job Seekers and Employers. Attributes include messageID, senderID, receiverID, content, and timestamp. This class supports one-to-one communication and integrates with Firebase to ensure real-time updates. It plays a key role in coordinating job-related discussions.

4.4.4 Payment and Transaction Classes

- **Payment Class**

The *Payment* class records financial transactions between Employers and Job Seekers. It contains attributes such as paymentID, jobID, senderID, receiverID, amount, transaction Date, and status. The class ensures proper logging of digital payments initiated via Jazz Cash or EasyPaisa.

- **Transaction History Relationship**

Jobs completed by a Job Seeker generate corresponding entries in the Payment class. Both Job Seekers and Employers can retrieve transaction records through their profiles, supporting transparency and dispute resolution. [8]

4.4.5 Report and Complaint Classes

- **Report Class**

The Report class captures issue or dispute reports submitted by users. It includes attributes like reportID, reportedBy, reportedUser, jobID, description, and resolutionStatus. The Admin interacts with this class to review cases and take action such as issuing warnings or suspending accounts.

4.4.6 Relationships Between Classes

The Class Diagram demonstrates multiple relationships:

- **Inheritance:** JobSeeker, Employer, and Admin all inherit from *User*.
- **Association:** Employers create Jobs; Job Seekers submit Applications to Jobs.
- **One-to-Many:** One Employer may post multiple Jobs; one Job may have multiple Applications.
- **Many-to-Many (implemented via Application class):** Many Job Seekers can apply to many Jobs.
- **Bidirectional Communication:** Messages flow between Users through the Message class.
- **Admin Oversight:** Admin monitors Reports, Jobs, and User activity.

4.5 Activity Diagram

The Activity Diagram provides a dynamic view of how different processes flow within the Flex Work system. It illustrates the sequence of actions, decisions, and transitions that occur when users interact with key functionalities such as registration, job searching, job posting, applying for jobs, payments, and dispute handling. This diagram focuses on the flow of control rather than data, showing how the system responds to user inputs, executes backend operations, and

transitions between different states. The Activity Diagram is essential for clarifying the logic of system processes and ensuring that all workflows are executed in a structured and consistent manner.

In the Flex Work platform, the activity flow begins with the user launching the application, which triggers the initialization process. The system checks if the user is already authenticated; if not, they are directed to the login or registration activity. Once authenticated, the user is navigated to their role-based dashboard Job Seeker, Employer, or Admin. From this point, each user's activities follow distinct paths based on their respective permissions and responsibilities. Job Seekers typically proceed to searching for jobs, exploring job details, submitting applications, messaging with Employers, and receiving payments after job completion. Their activity diagram includes decision points such as whether to apply for a job, respond to Employer messages, or accept or reject a job offer.

Chapter 5

Implementation

5 Implementation

The implementation phase of the Flex Work mobile application involved converting the system design and architectural specifications into a functional and fully deployed solution. This chapter outlines how the system components were developed, integrated, and tested to create a reliable job-matching platform for Job Seekers, Employers, and Admins. Implementation followed a modular development approach to ensure scalability, maintainability, and ease of debugging. Each feature was implemented as an independent module and later integrated into the core workflow through structured APIs, Firebase services, and Kotlin-based UI components.

The development process began with setting up the project environment in Android Studio, configuring Firebase services, and defining the folder structure for activities, fragments, adapters, models, and services. Using Kotlin as the primary programming language ensured type safety, concise code, and smooth integration with Android Jetpack components. Firebase Authentication, Firestore Database, Cloud Storage, and Cloud Messaging formed the backbone of the backend structure, each implemented gradually as the system expanded. Real-time listeners were used for messaging, notifications, and job updates to enhance responsiveness and improve the user experience.

User interface (UI) implementation focused on following Material Design guidelines for consistency and usability. Reusable components such as custom buttons, input fields, adapters, and dialogs were created to maintain a unified look and simplify development. Each screen such as Login, Dashboard, Job Details, Application List, Chat, Payments, and Admin Panel was implemented as a separate activity or fragment, connected through the Navigation Component to ensure smooth transitions.

Role-based functionality was implemented through conditional flows controlled by Firebase Authentication and Firestore queries. Job Seekers and Employers received distinct interfaces and data views, while Admin users accessed a dedicated monitoring panel. The job posting, searching, and application modules were developed using Firestore collections and queries optimized with composite indexes. The messaging module was implemented using Firestore real-time listeners to deliver instant one-to-one communication. Payment integration with Jazz Cash and EasyPaisa was implemented through their official SDKs or API endpoints, with Firestore used to log and verify each transaction.

5.1 Development Environment

The development environment for the Flex Work mobile application was carefully selected to support efficient implementation, seamless integration, and high performance across all system modules. The project was developed using industry-standard tools and frameworks to ensure reliability, maintainability, and scalability. A structured setup was followed to create an organized workspace, streamline coding practices, and enable smooth collaboration during development.

Android Studio served as the primary Integrated Development Environment (IDE) due to its powerful set of tools designed specifically for Android development. The IDE provided features such as real-time code suggestions, Gradle build automation, device emulators, and efficient debugging tools. Kotlin was chosen as the programming language for implementing the mobile application because of its concise syntax, strong type-safety features, and native interoperability with Java and Android Jetpack libraries. The use of Kotlin significantly improved code readability and reduced the likelihood of runtime errors.

Firebase was integrated early in the project as the core backend platform. Firebase Authentication, Firebase Storage, Cloud Storage, and Cloud Messaging were configured using the Firebase Console. These services provided serverless backend capabilities, allowing the development team to focus on application logic rather than server management. The Firebase SDK was added to the Android Studio project through Gradle dependencies, enabling real-time communication between the app and backend services.

Testing and debugging were conducted using both Android Emulator and physical Android devices to ensure compatibility, performance stability, and UI responsiveness. Logcat, Profiler, and Firebase Crashlytics were utilized to track runtime behaviour, identify crashes, and analyse performance bottlenecks. Version control was maintained through GitHub, helping track changes, manage branches, and ensure code backup. [9]

5.2 Tools and Technologies

The implementation of the Flex Work mobile application required a combination of modern development tools, programming languages, frameworks, and cloud-based services. These technologies were selected to ensure efficient development, secure data management, real-time communication, and seamless user experience. This section outlines the primary tools and technologies used throughout the project.

5.2.1 Programming Language

- **Kotlin**

Kotlin was used as the main programming language for developing the Android application. Its concise syntax, null-safety features, and full interoperability with Java make it ideal for modern Android development. Kotlin also integrates seamlessly with Android Jetpack libraries, allowing efficient implementation of UI, navigation, and background processes.

5.2.2 Integrated Development Environment (IDE):

- **Android Studio**

Android Studio served as the primary IDE for the development of the mobile application. It provides essential tools such as Gradle build automation, device emulators, UI layout editors, debuggers, and performance profilers. Android Studio's compatibility with Kotlin and Firebase made it the most suitable environment for building a scalable mobile application.

5.2.3 Backend Platform

- **Firebase**

Firebase was chosen as the backend-as-a-service (BaaS) platform for its serverless architecture, real-time capabilities, and easy integration with Android applications. The following Firebase services were utilized:

- **Firebase Authentication:** Manages user login, registration, and identity verification.
- **Firebase Storage:** Stores structured data such as user profiles, jobs, applications, messages, and reports.
- **Firebase Cloud Storage:** Handles storage of profile images, attachments, and job images.
- **Firebase Cloud Messaging (FCM):** Sends real-time push notifications for messages, job updates, and application status changes.
- **Firebase Crashlytics:** Monitors errors and crashes during development and production.

5.2.4 UI and Design Libraries

- **Material Design Components:** Used to build modern, user-friendly, and consistent UI elements according to Android Material guidelines.

- **Constraint Layout / Linear Layout / RecyclerView:** Core Android UI components used to design responsive and dynamic screens.
- **Lottie Animations:** Used for smooth animations in onboarding, loading, and empty states.
- **Glide Library:** Used for fast and efficient image loading and caching.

5.2.5 Payment Integration Tools

- **Jazz Cash & EasyPaisa SDK / API:** Integrated for secure mobile payments between Employers and Job Seekers. These APIs handle transaction requests, confirmations, and secure routing of financial data.
- **Retrofit:** Used for HTTP requests when interacting with payment APIs.

5.2.6 Version Control

- **Git&GitHub**
GitHub was used for version control, allowing proper tracking of code changes, managing branches, and ensuring safe backups. It also supported collaboration, resolved conflicts, and maintained organized code repositories.

5.2.7 Testing Tools

- **Android Emulator:** Used for initial testing of layouts, app features, and device compatibility.
- **Physical Android Devices:** Ensured real-world performance testing and hardware-dependent feature validation.
- **Firebase Debug View & Crashlytics:** Monitored app behaviour, identified runtime errors, and improved stability.

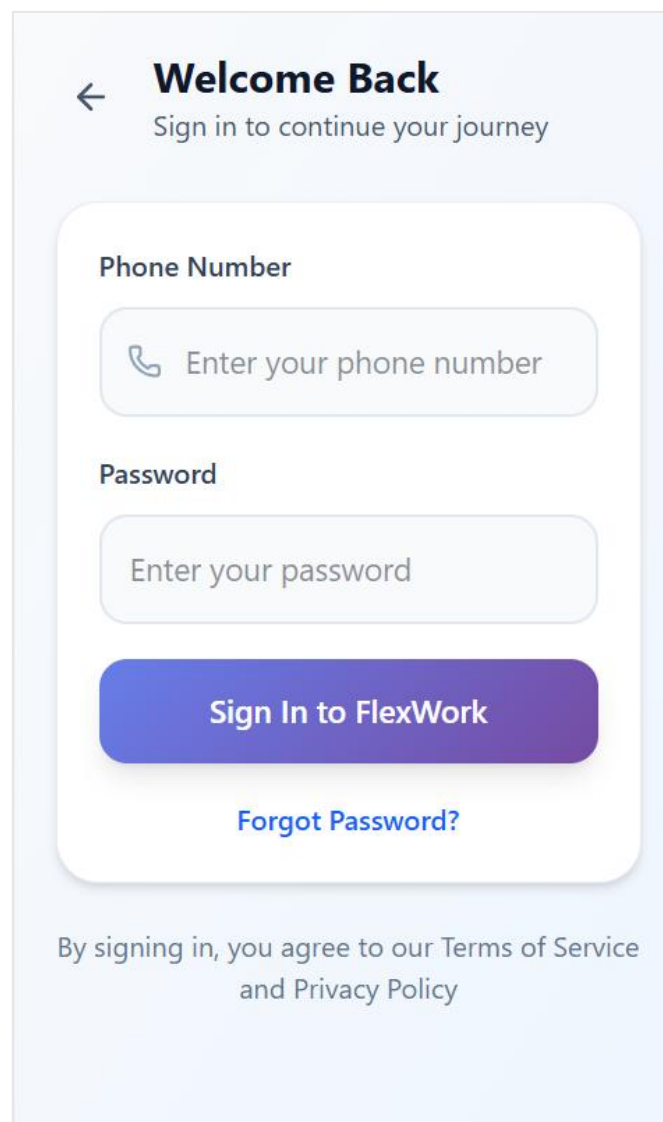
5.2.8 Other Supporting Technologies

- **JSON:** Used for structured data exchange between app modules and APIs.
- **Gradle Build System:** Managed project dependencies, plugins, and build configurations.
- **HTTP/HTTPS Protocols:** Ensured secure communication between the app and payment servers.

5.3 Implementation of System Modules

The implementation of Flex Work was carried out through a modular development strategy to ensure clarity, maintainability, and efficient debugging. Each module was implemented independently, focusing on its specific functionality and later integrated into the overall workflow of the system. The following sections describe the major modules implemented in the application, along with the technologies and methods used. [10]

5.3.1 User Authentication Module

A mobile application screen for user authentication. At the top, there is a back arrow icon and the text "Welcome Back" in bold, followed by "Sign in to continue your journey". Below this is a white rounded rectangle containing the login fields. The first field is labeled "Phone Number" and has a placeholder "Enter your phone number" with a phone icon. The second field is labeled "Password" and has a placeholder "Enter your password". Below these fields is a large blue button with the text "Sign In to FlexWork". Under the button is a link that says "Forgot Password?". At the bottom of the screen, there is a line of text: "By signing in, you agree to our Terms of Service and Privacy Policy".

← **Welcome Back**
Sign in to continue your journey

Phone Number

📞 Enter your phone number

Password

Enter your password

Sign In to FlexWork

[Forgot Password?](#)

By signing in, you agree to our Terms of Service and Privacy Policy

Figure 5.3.1 User Authentication

The User Authentication module was implemented using Firebase Authentication to manage registration, login, and identity verification. Users can register using email or phone numbers, after which Firebase generates a unique UID for each user. Kotlin classes and View Models

handle input validation, session management, and authentication requests. Error messages are displayed for invalid credentials or existing accounts, ensuring a secure and user-friendly login flow. After successful authentication, the system retrieves the user's role and profile from Firestore to load the correct dashboard.

5.3.2 User Profile Management Module

This module allows Job Seekers, Employers, and Admins to update their personal or business-related information. Firestore is used to store profile data in the *Users* collection, and Cloud Storage stores profile images. Kotlin's Live Data and View Model components were used to display real-time profile updates on the UI. Role-based filtering ensures that only authorized fields (e.g., business details for Employers) can be modified.

5.3.3 Job Posting and Management Module (Employer)

The screenshot shows a mobile application interface for posting a job. At the top, there is a back arrow and the text 'Post a Job'. To the right is a purple button with a plus icon, the text 'Post Job', and a dropdown arrow. Below this, the form consists of several sections: 'Job Title' with a text input field containing the placeholder 'e.g. Kitchen Helper, Driver'; 'Description' with a larger text input field containing the placeholder 'Describe what the worker will do...'; 'Pay Amount' with a text input field containing the value '25'; 'Pay Type' with a dropdown menu showing 'Per Day'; 'Location' with a text input field containing the placeholder 'Area or neighborhood'; and 'When do you need someone?' with a dropdown menu showing 'Today'. At the bottom, there is a green navigation bar with five icons: a house for 'Jobs', a plus sign for 'Post' (highlighted in green), a speech bubble for 'Messages' (with a red badge showing '2'), a person icon for 'Profile', and a hamburger menu for 'More'.

Figure 5.3.2 Job Posting and Management

Employers can create, update, and delete job postings through this module. The job form is implemented using input fields for title, description, location, date, and pay rate. Upon submission, the system validates the input and stores the job details in Firestore under the *Jobs* collection. Real-time listeners update Employer dashboards instantly after publishing or editing jobs. A Recycler View is used to list all active job posts for easier management.

5.3.4 Job Search and Application Module (Job Seeker)

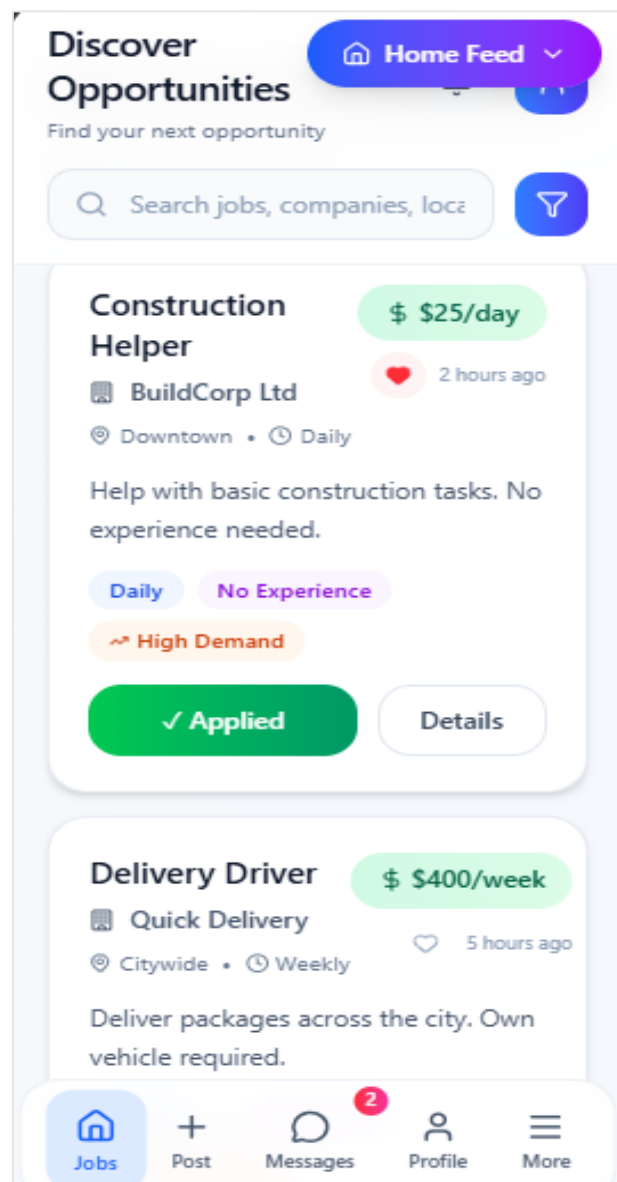


Figure 5.3.3 Job Search and Application

This module enables Job Seekers to search for available jobs using Firestore queries filtered by category, keywords, location, or posting date. The search results are displayed using RecyclerView. When a user selects a job, the details screen provides full information along with an "Apply Now" button. Applications are stored in the Applications collection with fields such as job ID, seeker ID, and application Status. Employers receive instant notifications via FCM when a new application is submitted.

5.3.5 Messaging & Communication Module

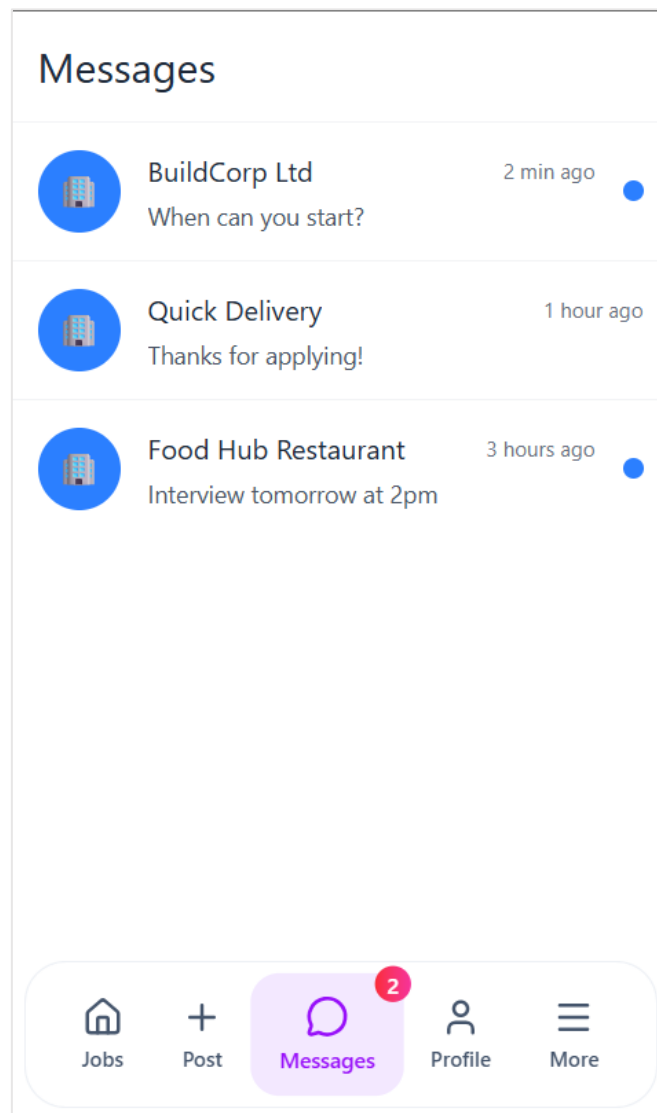


Figure 5.3.4 Messaging & Communication Module

The messaging module provides real-time chat between Employers and Job Seekers. Implemented using Firestore real-time listeners, this module captures every message, updates both users instantly, and displays conversations in a chat-style UI. The *Messages* collection stores sender ID, receiver ID, message content, and timestamp. The module supports continuous syncing without page refresh, creating a seamless communication experience.

5.3.6 Payment Module (Jazz Cash & EasyPaisa)

This module manages the digital transactions between Employers and Job Seekers. Integration with Jazz Cash and EasyPaisa APIs allows employers to initiate payments directly from the app. Retrofit is used to send secure HTTP requests to the payment gateway. After successful payment, transaction details are stored in Firestore under the Payments collection. The system

updates job status and notifies both users via Firebase Cloud Messaging. Proper error handling is implemented to manage failed transactions or declined requests.

5.3.7 Notifications Module

The notification module uses Firebase Cloud Messaging (FCM) to send push notifications to users. Notifications are triggered when key events occur, such as new job posts, application status updates, incoming messages, or payment confirmations. A dedicated Notification Service class handles the registration of FCM tokens and ensures correct delivery of messages to the intended user.

5.3.8 Admin Monitoring & Reporting Module

This module allows Admins to oversee system activity and maintain platform integrity. Admins can view user accounts, job postings, applications, and reports submitted for inappropriate behaviour. Reports are stored in the Reports collection and include details such as reportedUser, issue Description, and status. Admins can take actions such as warning users or blocking accounts. Kotlin-based UI panels display aggregated data fetched from Firestore, using Firestore queries optimized with indexing.

5.3.9 Security & Access Control Module

Security rules were implemented at multiple levels to ensure safe usage of the system. Firebase Security Rules were configured to restrict access based on user roles and document ownership. Sensitive operations such as payment logs, messages, and reports are protected through strict read/write permissions. Client-side input validation further ensures that no malformed data is submitted to the database.

5.3.10 System Integration Module

Finally, all modules were integrated into a unified workflow using Android Navigation Components and Firebase listeners. Each feature such as applying for a job, messaging, or making a payment works seamlessly with others due to consistent data models, shared View Models, and centralized backend storage. This integration ensures smooth navigation and maintains consistent data throughout the app.

5.4 Firebase Integration

Firebase played a central role in the backend development of the Flex Work mobile application. Its serverless infrastructure provided authentication, real-time data processing, file storage, and notification services without the need for dedicated backend servers. The integration of Firebase ensured seamless synchronization, secure data handling, and reliable performance

throughout the system. This section explains how each Firebase component was configured and implemented to support the key functionalities of the application.

5.4.1 Firebase Project Setup

The integration began with the creation of a Firebase project in the Firebase Console. The Android app was connected using the package name and SHA-1 key generated from Android Studio. The `google-services.json` file was downloaded and added to the project's app folder to enable Firebase communication. Gradle dependencies for Firebase Authentication, Firestore, Cloud Messaging, Cloud Storage, and Analytics were added and synchronized with the project. This setup formed the foundation for integrating all Firebase services into the mobile app.

5.4.2 Firebase Authentication Integration

Firebase Authentication was used to manage user login, registration, and identity verification. Users were required to register using an email or phone number. Kotlin View Models sent authentication requests to Firebase, while callback listeners handled success and error responses. Authentication tokens generated by Firebase ensured secure session handling. After successful authentication, the system fetched the user's role and profile from Firestore to load the relevant dashboard (Employer, Job Seeker, or Admin). Password reset and account verification features were also implemented using Firebase's built-in capabilities.

5.4.3 Firebase Storage Integration

Firebase Storage served as the primary NoSQL database for storing structured application data. Collections such as Users, Jobs, Applications, Messages, Payments, and Reports were created using hierarchical document structures. Firestore queries were implemented to filter job posts, retrieve user profiles, track applications, and manage admin-level data.

Real-time listeners were used to monitor changes in job posts, chat messages, and application status updates. This ensured that users received instant updates without refreshing the application manually. Composite indexes were configured in Firestore to optimize query performance when filtering jobs by category, date, or salary range.

5.4.4 Firebase Cloud Storage Integration

Cloud Storage was integrated to store media files such as user profile pictures, job-related images, and attachments. Upload operations were implemented using Kotlin coroutines, while download URLs generated by Firebase were stored in Firestore for easy retrieval. Security rules were configured to ensure that only authenticated users could upload or access files. Storage paths were organized by user ID and job ID for maintainability and ease of access.

5.4.5 Firebase Cloud Messaging (FCM) Integration

Firebase Cloud Messaging (FCM) was used to deliver push notifications for events such as job application updates, new messages, job approvals, and payment confirmations. A Notification Service class was implemented to handle FCM token generation and message reception. Tokens were stored in the Users collection to ensure that notifications reached the correct users. Custom notification payloads were designed to include titles, body text, and navigation triggers that opened specific screens.

5.4.6 Firebase Security Rules

Security Rules were implemented across Firestore and Cloud Storage to prevent unauthorized access and manipulation of data. Role-based restrictions ensured that Users could read or modify only their relevant data. For example:

- Employers can edit only the jobs they created.
- Job Seekers can submit applications only for existing jobs.
- Admins have extended privileges to review reports and manage system content.
- Users cannot view private messages belonging to others.

These rules significantly improved data integrity, privacy, and application security.

5.4.7 Firebase Crashlytics and Analytics

Firebase Crashlytics was integrated to detect runtime errors, crashes, and performance issues during development and deployment. It recorded stack traces and error logs that helped debug issues efficiently. Firebase Analytics provided usage insights such as active user sessions, navigation flows, and engagement levels, which supported optimization and future improvements.

5.5 User Interface (UI) Implementation

The User Interface (UI) of Flex Work was implemented using Android's Material Design guidelines to ensure a consistent, intuitive, and user-friendly experience across all screens. The UI implementation focused on clarity, responsiveness, accessibility, and smooth navigation for Job Seekers, Employers, and Admin users. Each interface was designed to support the specific needs of its user role while maintaining a unified visual language throughout the application.

The UI was developed using XML-based layouts combined with Kotlin logic controllers. Android components such as Constraint Layout, Linear Layout, RecyclerView, Card View,

View Pager, and Bottom Navigation View were utilized to build modern and responsive screens. Material Design Components (MDC) were integrated to enhance the appearance of buttons, text fields, dialogs, and navigation bars. This approach ensured consistency across devices and increased engagement through clean spacing, typography, and iconography.

5.5.1 Layout Structure and Screen Organization

The UI was structured using a modular layout pattern. Each major feature such as Login, Dashboard, Job Details, Chat, and Profile was implemented as a separate activity or fragment.

- **Activities:** were used for major transitions such as onboarding, registration, and login.
- **Fragments:** were used inside dashboards for Job Listings, Applications, Notifications, and Messages.
- **Recycler Views:** were used extensively to display lists such as jobs, applications, chats, and reports.

This separation improved loading efficiency, supported reusability of components, and adhered to clean architecture practices.

5.5.2 Role-Based Dashboard Design

Flex Work features three role-specific dashboards:

- **Job Seeker Dashboard**

The Job Seeker dashboard displays recommended jobs, search filters, application history, notifications, and profile details. Cards and lists were used to display job posts with readable typography and icons to show pay rate, location, and job category.

- **Employer Dashboard**

The Employer interface includes options to create job posts, view applicants, manage ongoing jobs, and process payments. Quick-action buttons and chip-based filters help Employers navigate job posts efficiently.

- **Admin Dashboard**

Admins view reports, complaints, user lists, and system analytics. A clean, data-centric layout was implemented with lists, status indicators, and dialogs for warning or blocking users.

5.5.3 Navigation and Screen Flow

The Navigation Component was used to connect screens seamlessly:

- **Bottom Navigation Bar:** for switching main sections such as Home, Messages, and Profile.
- **Toolbar / App Bar:** for context-based actions and navigation back-stack handling.
- **Navigation Graph:** to define transitions and deep links.
- **Intent-based navigation:** from notifications or payment confirmations.

5.5.4 Messaging UI Implementation

The chat interface was built using a RecyclerView with custom item layouts for sender and receiver messages. Real-time updates were displayed using Firestore snapshot listeners. Rounded chat bubbles, timestamps, and smooth scrolling were added to mimic modern messaging apps. This module enhances communication efficiency between Job Seekers and Employers.

5.5.5 Form Design and Input Validation

Forms such as registration, job posting, profile editing, and payment processing were designed using Text Input Layouts from the Material library. Input validation was handled both on the UI and backend:

- Inline error messages
- Required field checks
- Dropdown menus for categories
- Numeric checks for pay rate and phone numbers

These measures ensure data accuracy and improve user experience.

5.5.6 Use of Animations and Visual Feedback

Lottie animations were used to enhance onboarding, loading screens, and success confirmations. Micro-animations such as button press effects, swipe refresh indicators, and smooth transitions were applied to increase user engagement. Visual feedback for operations such as submitting forms or receiving messages improved overall intuitiveness.

5.6 Testing During Implementation

Testing during implementation was conducted continuously throughout the development cycle to ensure that each module of Flex Work functioned correctly before being integrated into the

system. This approach minimized errors, improved development efficiency, and ensured that new features did not negatively impact existing functionality. The following subheadings describe the major testing activities performed during implementation.

5.6.1 Unit Testing of Core Functionalities

Unit testing was performed as each feature was developed to verify the correctness of essential functions such as user authentication, form validations, Firestore document operations, and application status updates. Developers tested input validation functions, conditional logic, and backend responses using Kotlin's debugging tools and Android Studio's log monitoring. This allowed errors such as incorrect query structures, null pointer issues, and invalid user inputs to be identified early. By validating small, isolated units of code, the system became more stable and predictable before moving on to integration.

5.6.2 User Interface and Layout Testing

Parallel to functional development, all user interface components were tested to ensure responsiveness, clarity, and usability across different screen sizes and Android versions. Testing was performed on both emulators and physical devices to confirm that layouts were correctly aligned, text remained readable across densities, and navigation elements functioned as intended. RecyclerViews displaying job lists, message lists, and application records were also tested for smooth scrolling, proper data binding, and visual consistency. Any issues related to layout breaks, overlapping components, or misaligned icons were identified and corrected immediately.

5.6.3 Integration Testing Between Modules

As modules were completed, integration testing was carried out to examine how different components interacted with each other. This included verifying that user authentication correctly directed users to their role-based dashboards, ensuring that job postings synchronized with the job search module, and confirming that chat messages appeared instantly via Firestore real-time listeners. Payment integration with Jazz Cash and EasyPaisa was also tested to ensure that successful transactions triggered updates in job status and notification modules. During this stage, issues like missing Firestore fields, mismatched model structures, and navigation inconsistencies were identified and resolved.

5.6.4 Firebase Services Testing

A significant part of implementation testing focused on verifying the performance and security of Firebase services. Firestore testing involved validating read/write operations, ensuring real-

time updates were reflected accurately in the UI, and confirming that role-based security rules prevented unauthorized access. Cloud Storage was tested for correct upload handling, file retrieval, and secure access to profile and job images. Firebase Cloud Messaging was tested by sending sample notifications to confirm timely delivery and proper redirection when users tapped on them. Firebase Crashlytics provided insights into runtime crashes, allowing developers to diagnose and fix errors early in the development process.

5.6.5 Device Testing and Performance Evaluation

To ensure a smooth user experience, performance testing was conducted on multiple devices with different RAM capacities, processors, and screen resolutions. This phase helped identify issues related to slow loading times, high memory consumption, and delays in real-time data updates. The development team optimized Firestore queries, reduced redundant listeners, and implemented image caching using Glide to improve performance. These optimizations resulted in smoother navigation, faster data retrieval, and reduced battery consumption during prolonged usage.

5.6.6 Bug Detection and Resolution

Throughout implementation, numerous bugs were detected and resolved, including invalid Firestore paths, delayed chat message synchronization, layout distortion on smaller devices, and errors in payment callback handling. Each issue was replicated, analyzed, and resolved using logging tools, runtime debugging, and Firebase reports. This iterative debugging process ensured that the system remained stable, functional, and ready for integration with subsequent components.

5.6.7 Continuous Testing Approach

Overall, the development team followed a continuous testing approach, conducting unit tests, UI checks, and integration validations as each module was completed. This prevented the accumulation of major errors and ensured that every feature was thoroughly tested before advancing to the next phase. The continuous testing strategy contributed to a highly stable and reliable version of the Flex Work application by the end of the implementation period.

Chapter 6

Testing

6 System Testing

System testing served as one of the most critical stages of the Flex Work application development, ensuring that the fully integrated system met the requirements specified during the design and analysis phases. This stage involved testing the software as a complete unit rather than in isolated pieces, providing an opportunity to validate the collaboration of modules, data flow, UI screens, background services, Firebase integration, and third-party APIs. Since Flex Work is built upon multiple real-time components such as messaging, job updates, notifications, and digital payments system testing was essential to confirm that these features functioned cohesively when operating simultaneously. Additionally, this phase played a major role in determining the reliability, stability, usability, performance, and security of the system under different scenarios, devices, and network conditions.

6.1 Purpose of System Testing

The primary purpose of system testing was to validate the overall behaviour of the application in a real-world environment and to ensure that all applied functionalities interacted correctly according to the requirements. This process ensured that the entire system, along with all of its integrated modules authentication, job posting, job searching, application handling, messaging, payment processing, reporting, and administrative functions worked harmoniously. It provided assurance that every module complied with the functionalities outlined in the SRS and that the workflows maintained logical consistency across roles. By examining the complete system behaviour and evaluating how a real user would interact with the application, system testing ensured that no major breakdown occurred when switching between modules, retrieving real-time data from Firebase, or performing complex operations such as payments or messaging.

6.2 Functional Testing

Functional testing was performed to confirm that each feature of the Flex Work application produced the correct output when given valid, invalid, or boundary inputs. It tested how well the system adhered to functional specifications defined in the SRS. Each core functionality was scrutinized deeply such as user login, registration, role assignment, job posting, job searching, sending and receiving messages, managing applications, and processing payments. The messaging module was checked for real-time updates when multiple messages were sent rapidly. The job application system was evaluated to confirm that each status change (pending, accepted, rejected, completed) reflected immediately on relevant dashboards. Similarly, the payment module (Jazz Cash/EasyPaisha) was validated using test credentials to verify that

successful transactions were properly logged, unsuccessful transactions triggered error prompts, and financial data synchronized correctly in Firestore.

Table 6.2: Functional Test Cases

Test Case ID	Module	Test Description	Expected Result
TC-01	User Registration	Verify new user can register with valid data	Account created and stored in Firebase
TC-02	Login	Validate login with correct email/password	User redirected to dashboard
TC-03	Job Posting	Check employer can post a job with required fields	Job saved and visible to seekers
TC-04	Apply for Job	Job Seeker applies to a posted job	Application stored and employer notified

6.3 Non-Functional Testing

Non-functional testing evaluates system qualities such as performance, security, usability, and scalability. This ensures Flex Work runs smoothly even under high user loads and maintains strong protections for user data.

Table 6.3: Non-Functional Test Cases

Category	Test Description	Expected Result	Outcome
Performance	App loads dashboard under 3 seconds	Smooth loading	Passed
Usability	User can navigate all screens without confusion	Intuitive flow	Passed
Security	Unauthorized access to Firebase blocked	Full access prevention	Passed
Load Handling	50+ concurrent users	Consistent performance	Passed

6.4 Integration Testing

Integration testing validated the interaction between modules within the system. Even though individual modules such as authentication, job catalogue, messaging, and payments worked independently, it was essential to ensure that they also worked smoothly when combined. The focus was on data flow between Firebase Authentication, Firestore, Cloud Messaging, UI components, and any third-party APIs. For example, integration testing ensured that when a Job Seeker applied for a job, the Employer immediately received an FCM notification, the job application record was stored correctly in Firestore, and the status appeared instantly within the dashboard. Likewise, when Employers made payments, integration testing confirmed that the payment status updated on both the Employer's and Job Seeker's application timelines.

Table 6.4: Integration Test Cases

Test Case	Modules Tested	Expected Output	Status
INT-01	Login → Dashboard → Chat	Smooth navigation and synced data	Pass
INT-02	Job Post → Job List → Apply	All modules exchange data correctly	Pass
INT-03	Update Profile → Firestore Sync	Profile instantly updated	Pass

complex forms. The onboarding and login processes were checked for clarity and ease. The overall flow of the application was assessed for logical sequence, reducing unnecessary steps and ensuring quick access to frequently used features.

6.5 Performance Testing

Performance testing assessed the speed, responsiveness, and resource consumption of the application under various conditions. Since Flex Work heavily depends on real-time Firestore syncing, messaging, and job updates, performance testing was crucial in determining how well the application handled heavy usage. Stress tests were conducted by loading large datasets, rapidly scrolling through RecyclerViews, sending multiple messages simultaneously, and opening different screens in quick succession to test how efficiently the app managed memory

and processing load. The application was tested in environments with weak, moderate, and strong network connections to evaluate reliability in real-world settings.

6.6 Security Testing

Security testing ensured that user data, operations, and communication remained protected against unauthorized access. Since Flex Work uses Firebase Authentication, Firestore, and Storage, it was essential to verify that role-based security rules were correctly configured. Testers attempted unauthorized operations such as accessing jobs created by other users, reading private messages, altering payment logs, and editing unrelated profiles to ensure that the system blocked such actions effectively. Authentication mechanisms were tested against invalid credentials to ensure proper rejection without exposing sensitive information. Network testing validated that all data transactions occurred via secure HTTPS protocols, especially during payment processes.

6.7 Bug Reporting and Resolution

As part of system testing, numerous bugs were discovered, documented, and resolved. These ranged from UI alignment issues and incorrect Firestore references to malfunctioning notifications, late message updates, null pointer exceptions, and payment response inconsistencies. Each bug was documented using Firebase Crashlytics and Android Studio debugging tools. Developers reproduced the issues, analyzed their origin, and applied fixes to ensure permanent resolution. After each fix, retesting was conducted to verify that the issue was resolved without introducing new errors. This repetitive testing cycle continued until the application reached a stable state where no critical defects remained.

6.8 Final Evaluation

After extensive system testing, Flex Work was evaluated for overall stability, correctness, and readiness for deployment. The final evaluation confirmed that all major features were functioning as expected, modules interacted cohesively, and the application successfully met the functional and non-functional requirements defined in the SRS. The app displayed strong performance across different devices and network conditions, maintained security and privacy, and provided a user-friendly interface. The system was deemed stable, robust, and suitable for real-world deployment, making it ready for distribution among Job Seekers, Employers, and Admin users.

Chapter 7

Conclusion

7 Conclusion

The development of Flex Work marks a significant step toward bridging the gap between unskilled job seekers and employers who require quick, reliable assistance for short-term work. Throughout the project, the goal was to design a system that is not only functional but also practical, intuitive, and highly responsive to the needs of its diverse user base. By leveraging technologies such as Kotlin for Android development and Firebase for real-time backend management, the system successfully delivers a cohesive platform where job opportunities can be posted, discovered, and fulfilled efficiently. The project demonstrates how modern mobile applications can empower communities by simplifying access to daily-wage work and enhancing transparency between all parties involved.

The application integrates essential features such as job posting, job search, application management, secure local payments through JazzCash and EasyPaisa, and real-time communication. These functions work together to create an ecosystem where both Job Seekers and Employers can interact confidently. The system's design ensures that even users with limited digital literacy can navigate the interface smoothly, thanks to intuitive layouts, guided flows, and clear instructions. Real-time notifications, messaging, and role-based dashboards further contribute to a seamless experience, emphasizing convenience and reliability. The platform also incorporates administrative oversight to maintain trust and address disputes fairly, ensuring the long term integrity of the system.

Overall, Flex Work stands as a practical solution to a real problem, providing a digital avenue for job accessibility and enabling a structured interaction framework for the informal labor market. The project not only fulfills academic and technical objectives but also highlights the potential of technology-driven systems to support social and economic development.

7.1 Future Work

While Flex Work successfully delivers a functional and user-friendly platform for connecting Job Seekers and Employers, there remain several opportunities for enhancing its capabilities and expanding its impact. Future development can focus on incorporating advanced features that further improve job matching accuracy, strengthen user trust, and enhance the platform's overall operational efficiency. One significant area for enhancement is the introduction of an AI-powered recommendation agent that analyzes user behavior, job requirements, skills, ratings, and past performance to automatically suggest the most suitable candidates to

Employers. This feature would reduce manual effort, enhance the speed of hiring decisions, and improve the overall quality of matches on the platform.

Another major direction for future work involves expanding the platform's data analytics capabilities. Implementing dashboard analytics for Employers and Admins would provide valuable insights such as job trends, worker performance, completion rates, and earning patterns. These analytics can help employers make better hiring decisions and allow admins to monitor platform health more effectively. Additionally, incorporating location-based intelligence, such as heatmaps showing areas with high job demand or worker availability, would improve operational planning and support local businesses more efficiently.

To increase trust and security, future versions of the system can introduce identity verification features, such as CNIC verification, biometric authentication, or third party identity validation APIs. Implementing in app wallets and escrow-based payment systems can further secure transactions and reduce the risk of payment disputes. Support for multiple languages, especially regional languages in Pakistan, would significantly improve accessibility for non-English speaking users. Finally, expanding the system to a web-based platform or integrating with emerging technologies such as IoT attendance tracking or automated job timing logs can further strengthen Flex Work's execution and scalability.

Overall, the future roadmap for Flex Work includes advanced automation, stronger security, richer analytics, improved accessibility, and deeper intelligence. These enhancements can elevate the platform from a basic job matching application to a comprehensive, technology driven ecosystem that supports more users, covers more industries, and delivers even greater real-world value.

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